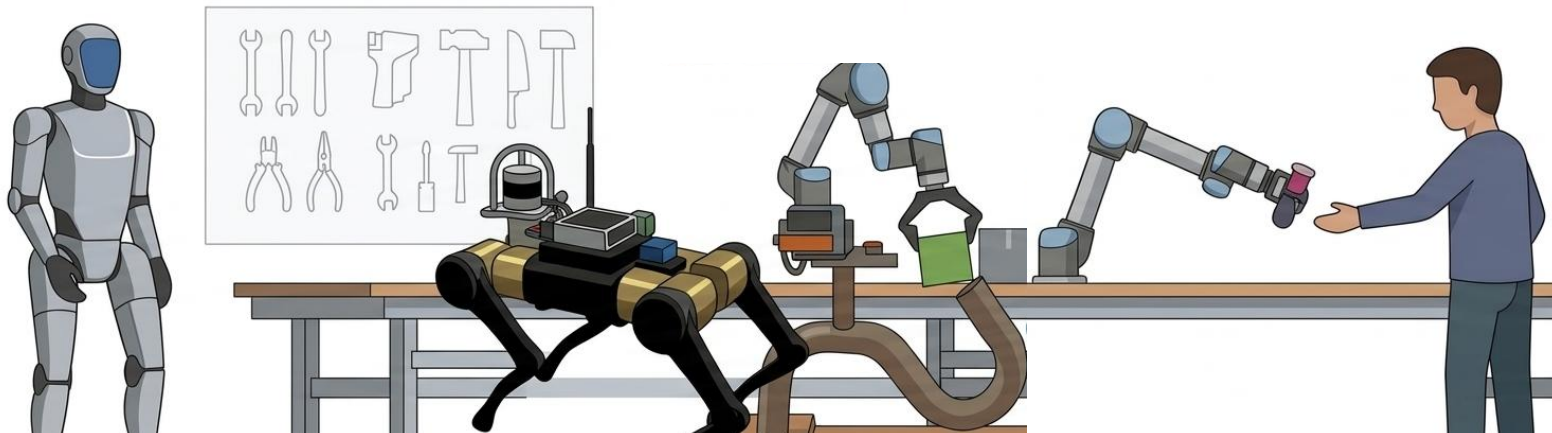


# Robust Intelligence 강인지능 Robotics Lab 로봇틱스 랩

Research Highlight



Dr. Daehyung Park  
Associate Professor  
School of Computing, KAIST



**Name** Daehyung Park

**Affiliation** School of Computing, KAIST

**Degree** Robotics Ph.D., Georgia Tech.

**Research Interest** Robotics, LfD, TAMP, NLG/NLU

## Major Research Experience & Awards

- ✓ Postdoctoral Associate, MIT CSAIL (2018-2020)
- ✓ Research Engineer, Samsung (2008-2012)
- ✓ ICRA Outstanding Planning Paper Award (2023)
- ✓ ICRA Outstanding Navigation Paper Award Finalist (2022)
- ✓ Google Research Scholar Award (2022)



## Representative Work



## Professional Activities

- ✓ Director, Korean Robotics Society (2025~Present)
- ✓ Demo Chair, CoRL (2025)
- ✓ Program Chair, RiTA (2024)
- ✓ Organizer, Workshop at Humanoids (2024)
- ✓ Organizer, Workshop at RSS (2023)

## Funding Agencies



**Opened**  
in 1971

**Located**  
in Daejeon, S. Korea



**Students**  
12,348

**Comprise of 6 Colleges**  
7 Schools  
33 Departments  
55 Programs

**Faculty and Staff**

Faculty 731

Staff 907

**International Population**

Faculty 137

Students 929



# Robust Intelligence & Robotics (RIRO) Lab

**Opened**  
in 2020 Fall

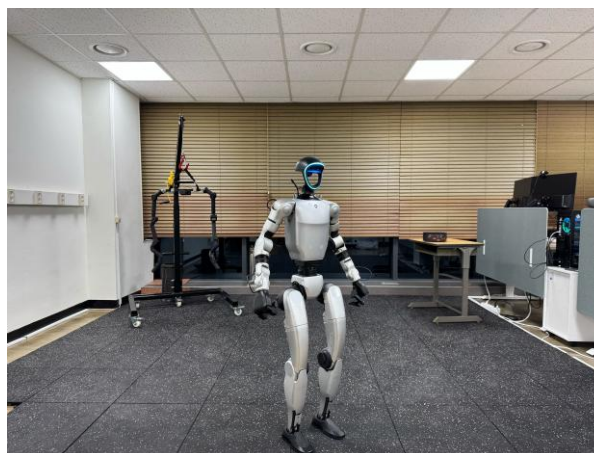
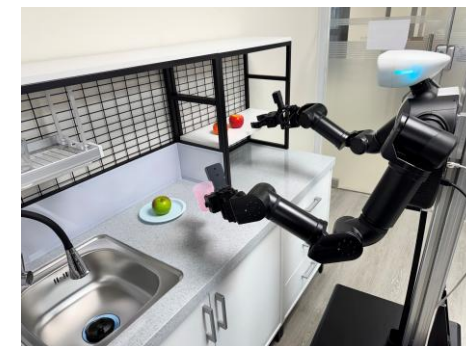
**Affiliation**  
School of Computing

**Members**  
2 Post Doc.  
5 Ph.D.  
12 M.S.

**Research Area**  
Manipulation,  
Skill Learning,  
HRI



KAIST



Robust Intelligence & Robotics Lab



# Robust Intelligence & Robotics (RIRO) Lab

## Projects



**Gov.  
Proj**

9



**Industry  
Proj**

1

## Facilities



**Office  
room**

3



**Robot  
Lab**

3



**Server  
Workstation**

7ea.  
8ea.



**3D  
Printer**

2ea.

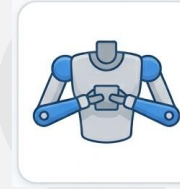


## Equipments



**Humanoid  
Robot**

1ea.



**Dual  
Arm**

1ea.



**Robot  
Arm**

2ea.



**Mobile  
Arm**

2ea.



**Robot  
Hand**

2ea.

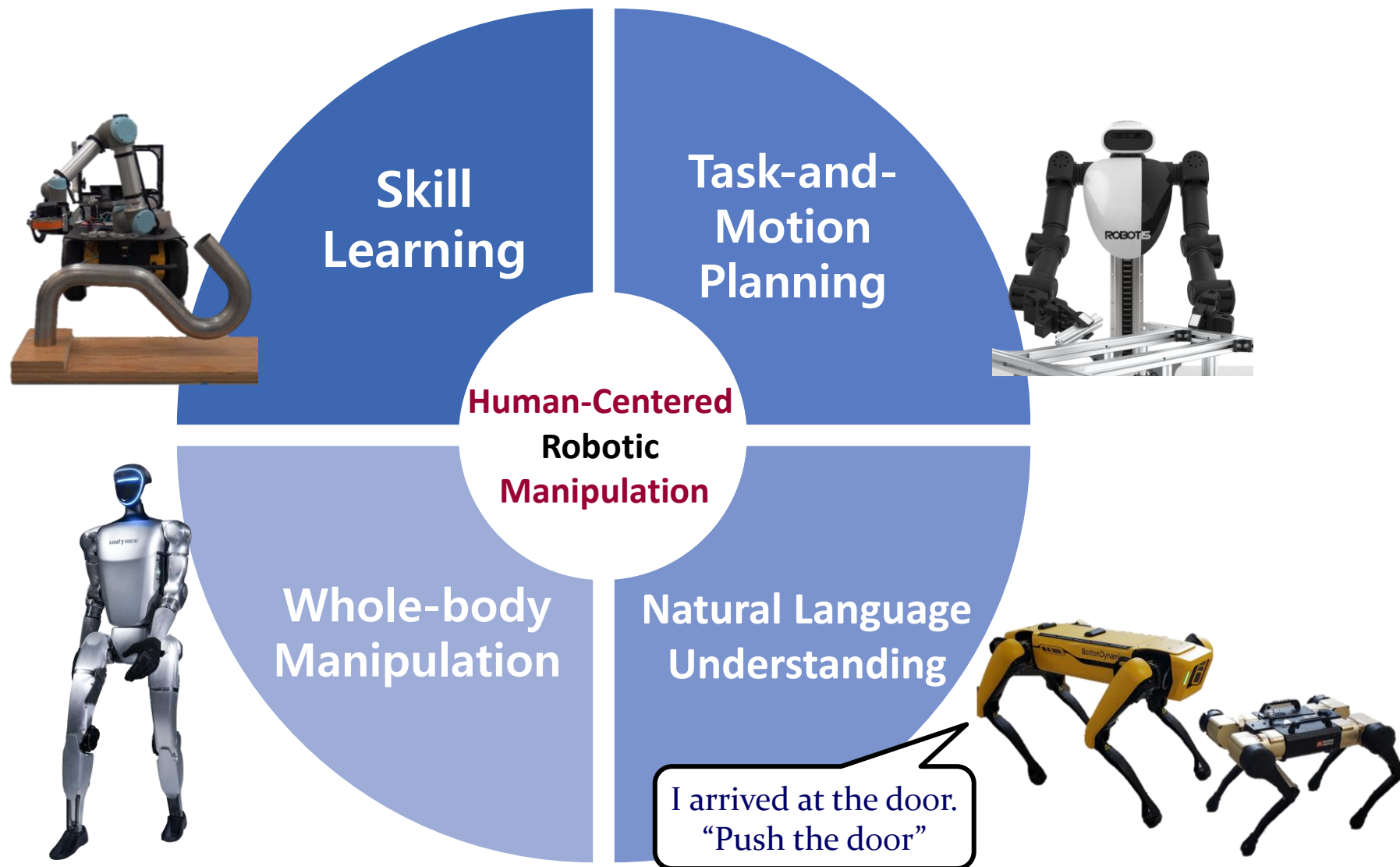


**Quadruped  
Robot**

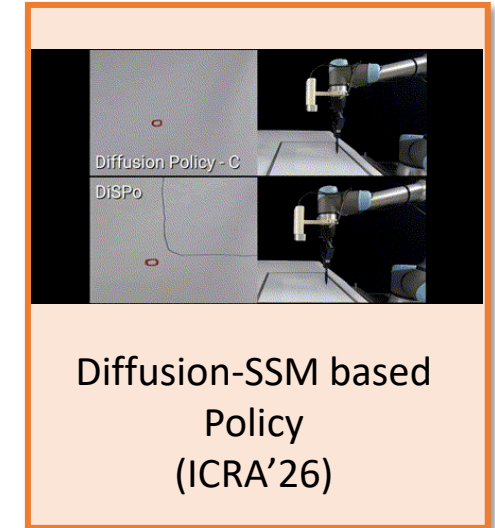
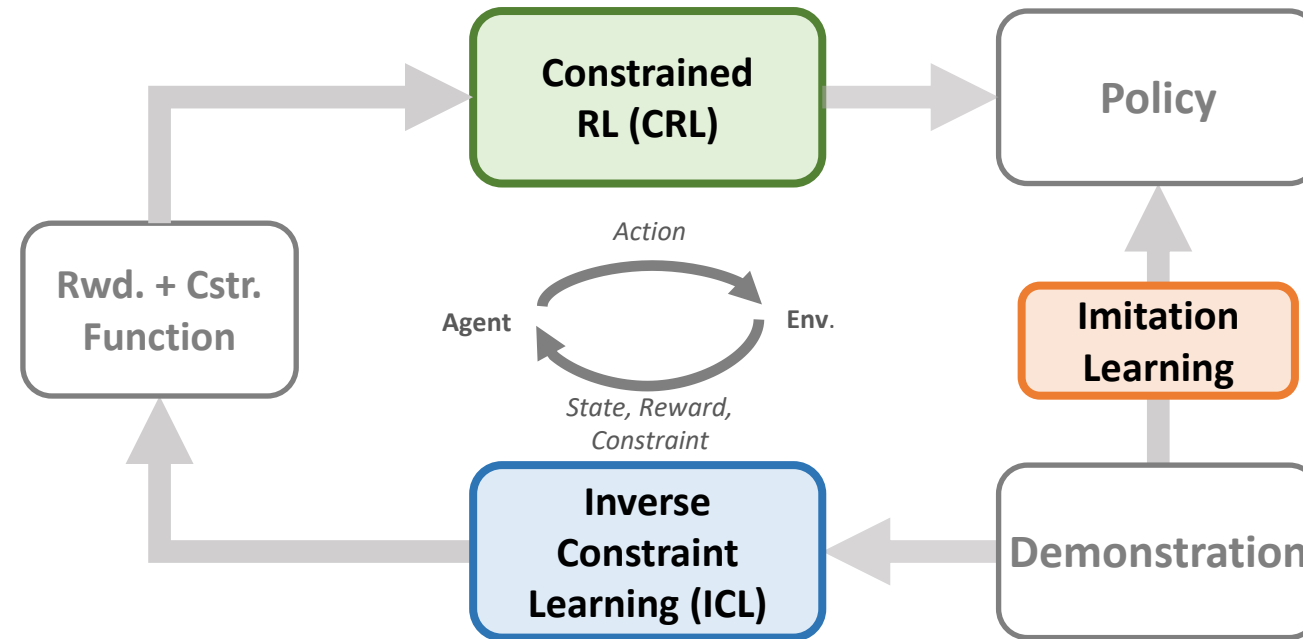
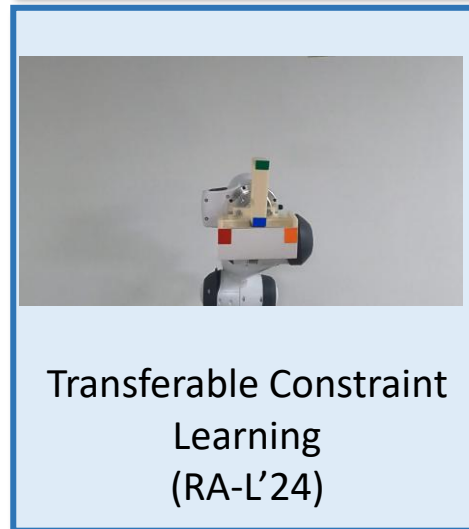
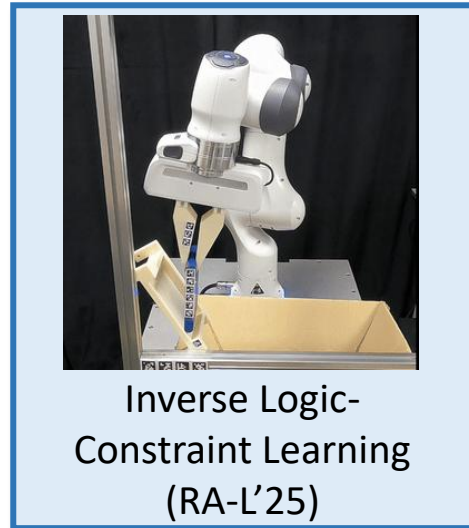
1ea.

# Human-Centered Robotics:

## How to bridge the gap between humans and robots?



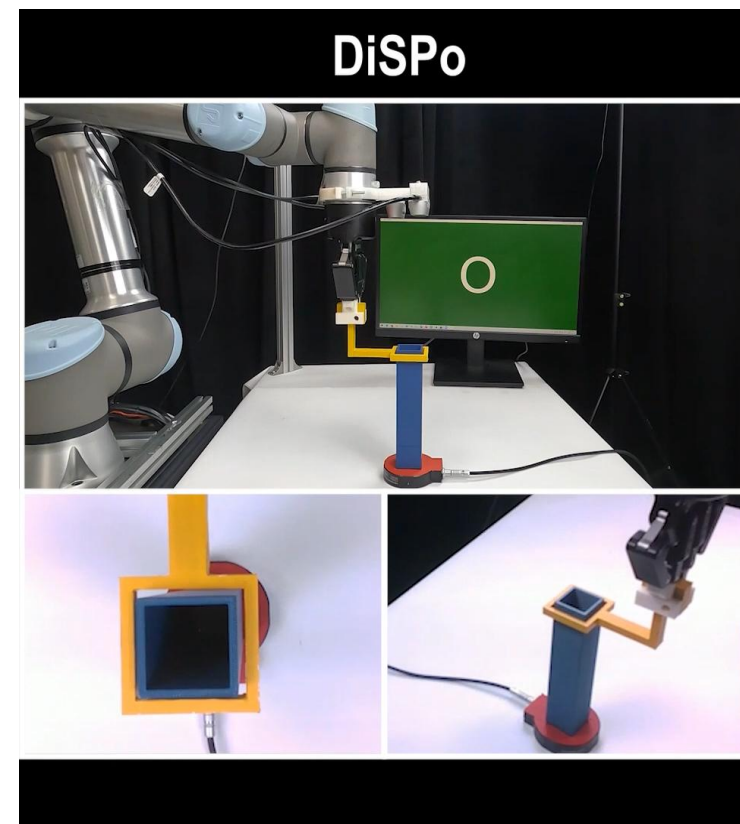
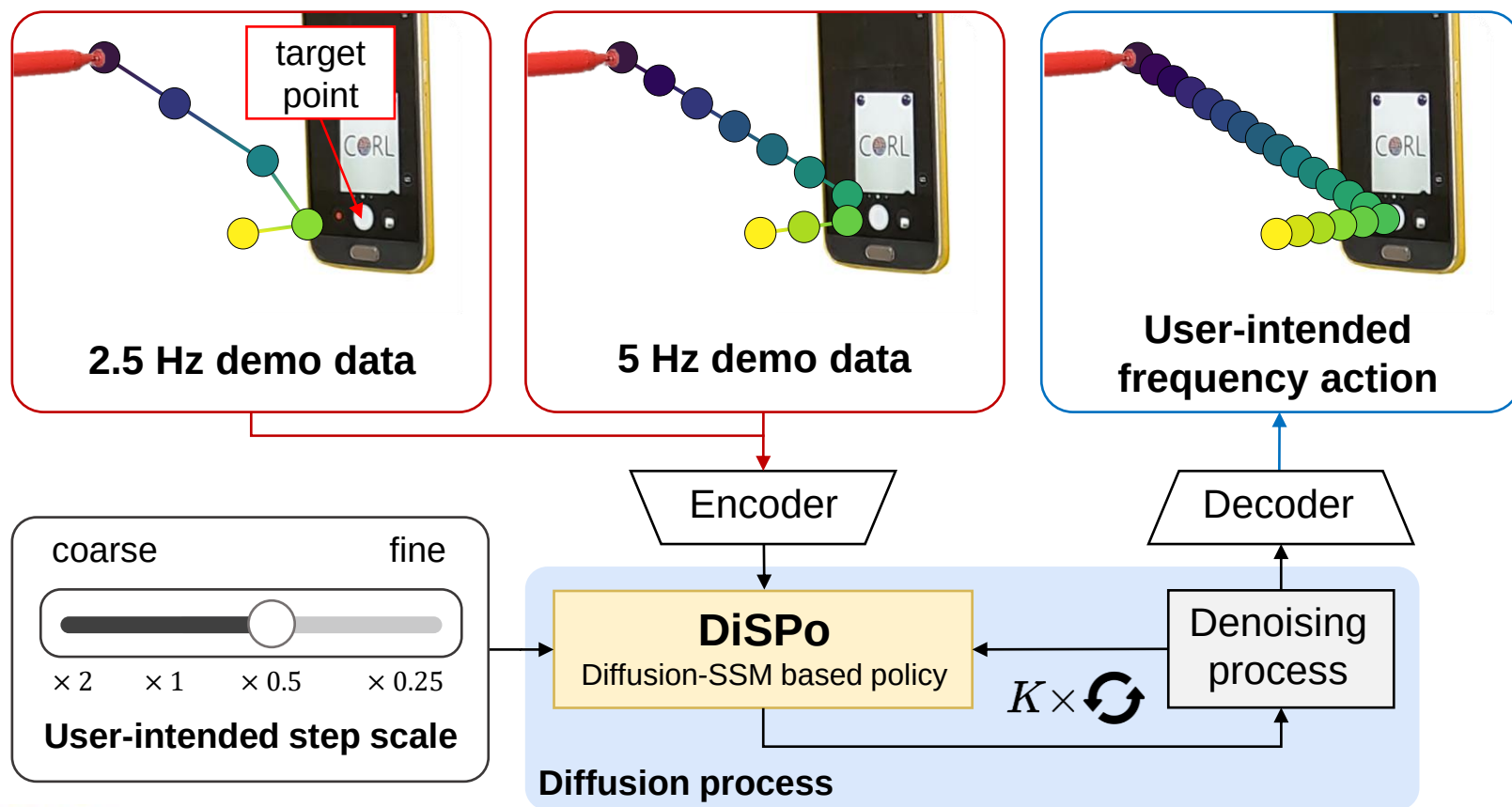
# Inverse Skill Learning



# Coarse-to-fine manipulation

## DiSPo: Diffusion-SSM based Policy

DiSPo that leverages a **state-space model**, Mamba, to learn from diverse coarse demonstrations and generate **multi-scale actions**.



# Small Vision-Language Action Model

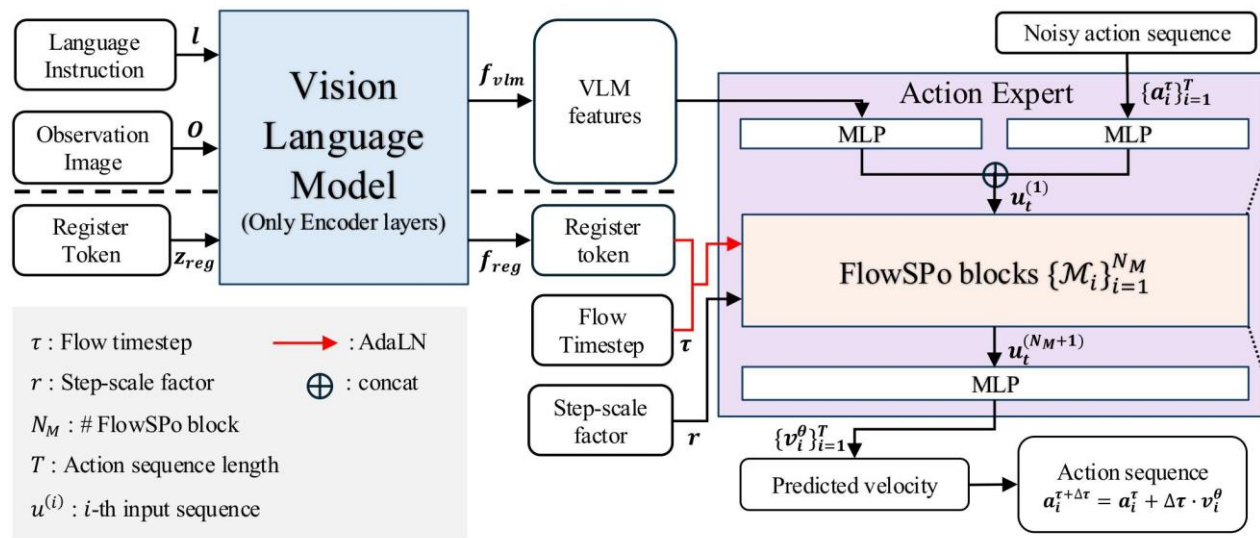
## FloSPo-VLA (1B)



Sangyun

(to be submitted)

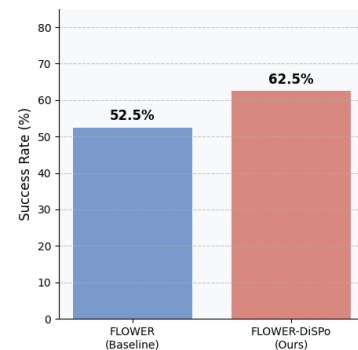
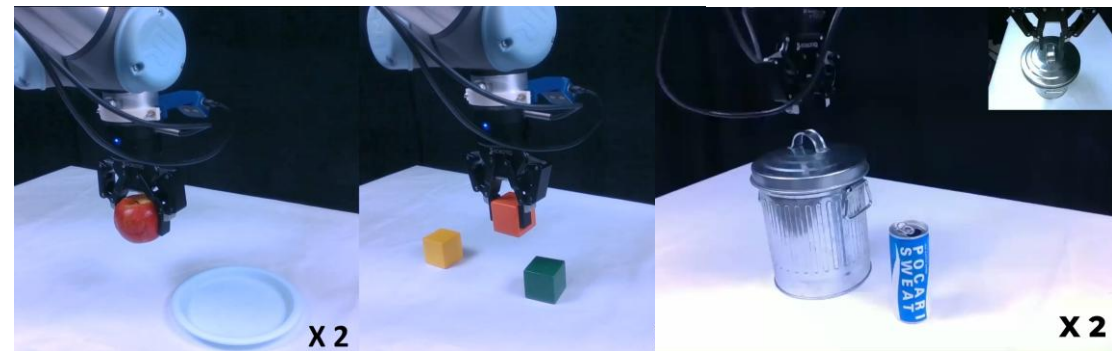
**Multi-granularity learning** effectively learns multi-frequency/downsampled dataset.



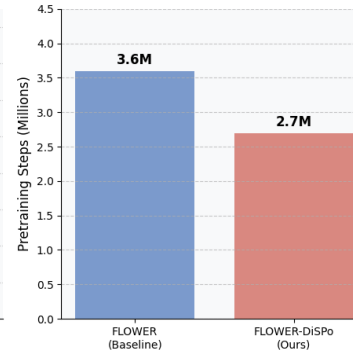
(FloSPo-VLA~1B)

**Self-attention replacement**

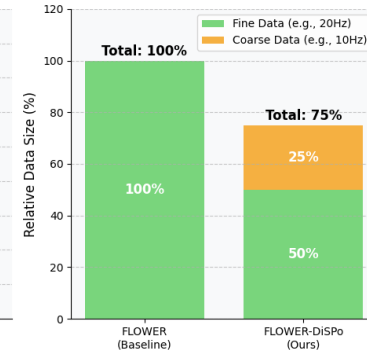
$$O(N^2) \rightarrow O(N)$$



**Success Rates**



**Pretraining steps**



**Relative dataset size**

**time & storage complexities ↓**



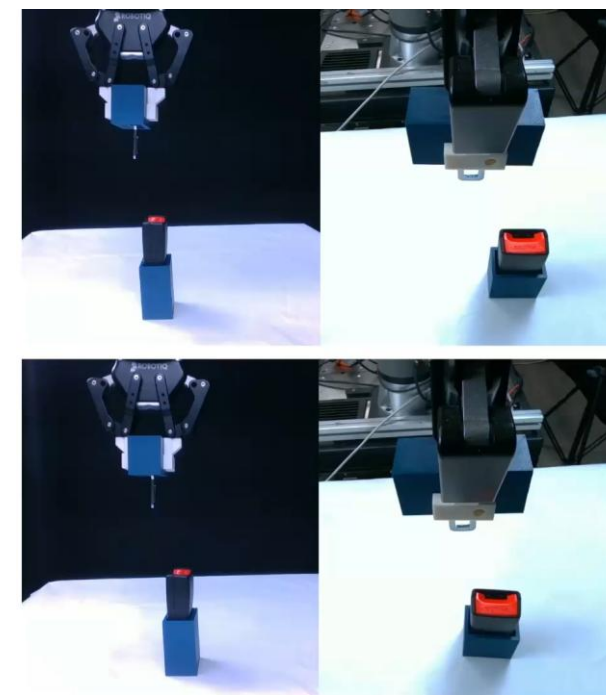
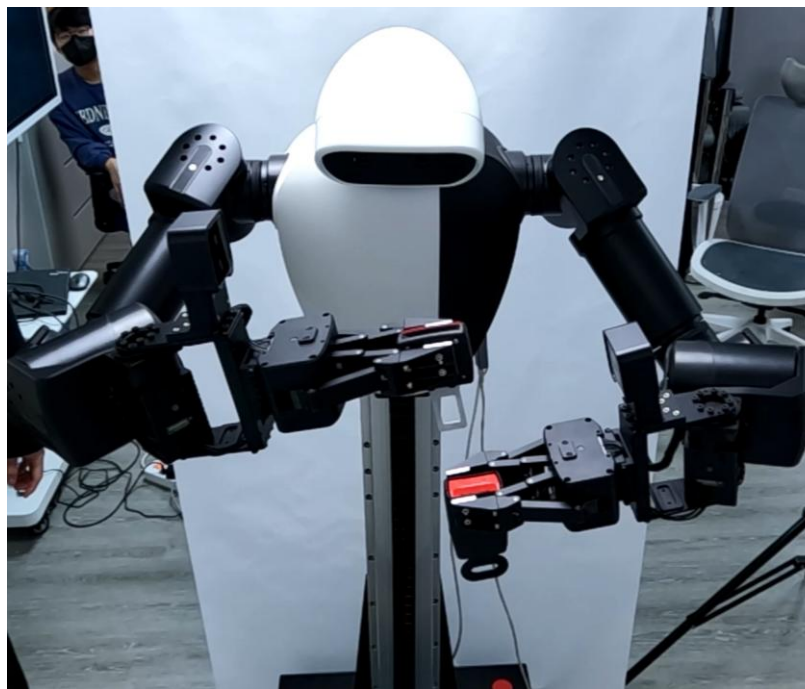
Jun



Jaehyeong

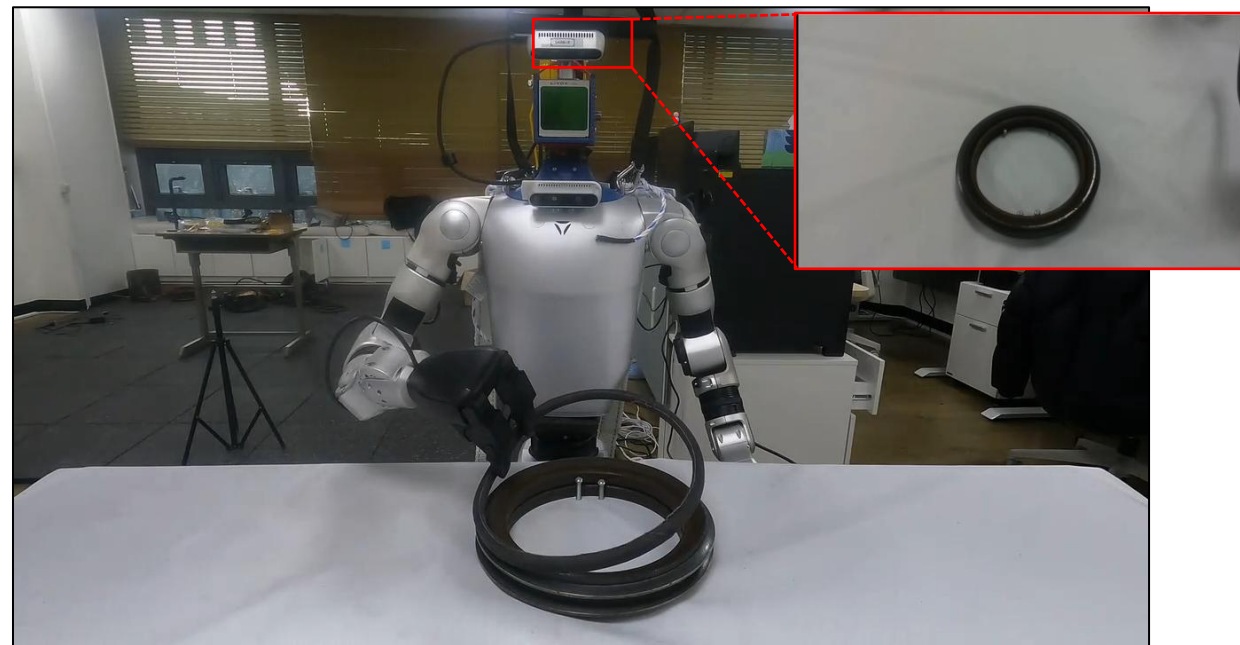
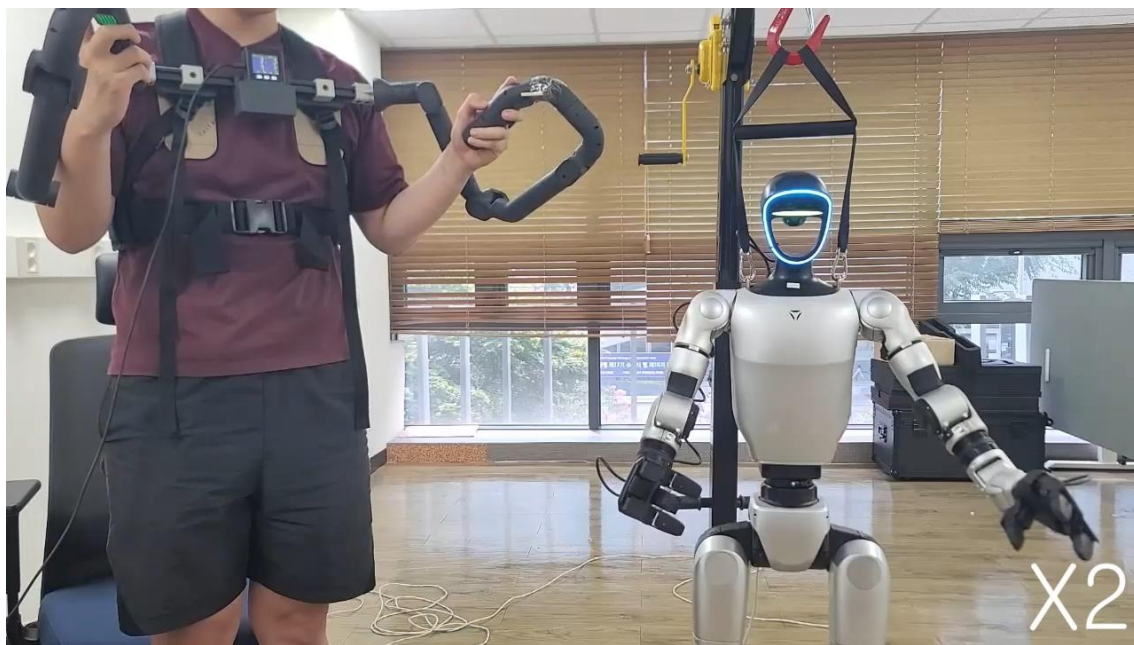
# Humanoid Precision Task Execution

We develop a behavior tree–based system for executing precision tasks on humanoid robots.



# Humanoid Bi-manual Manipulation

We learn manipulation skills for highly elastic deformable objects using a humanoid's dual arms via reinforcement learning.



# Implicit Neural-Representation Learning for Elastic Deformable-Object Manipulations

SAMSUNG



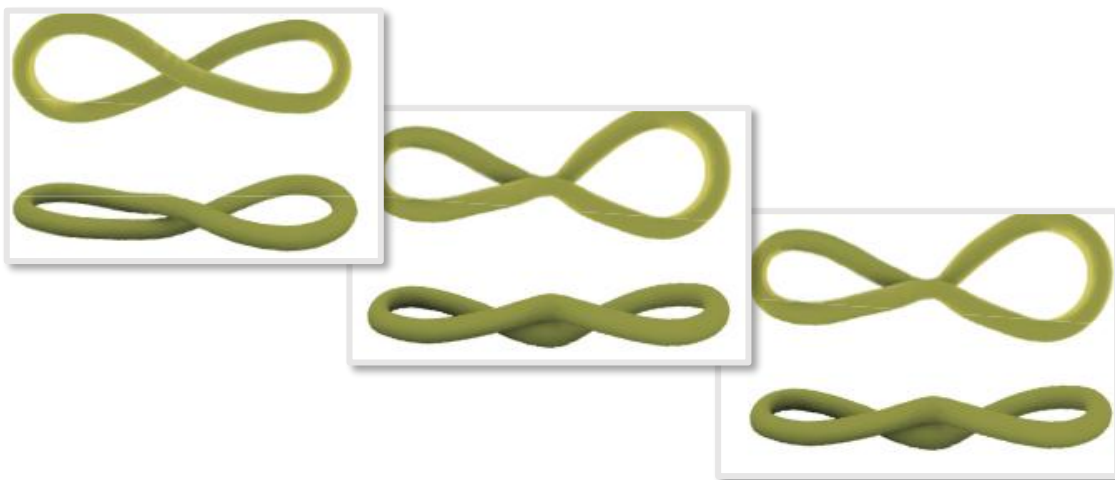
Minseok

15

**Problem:** Deformable elastic-object manipulation in real world

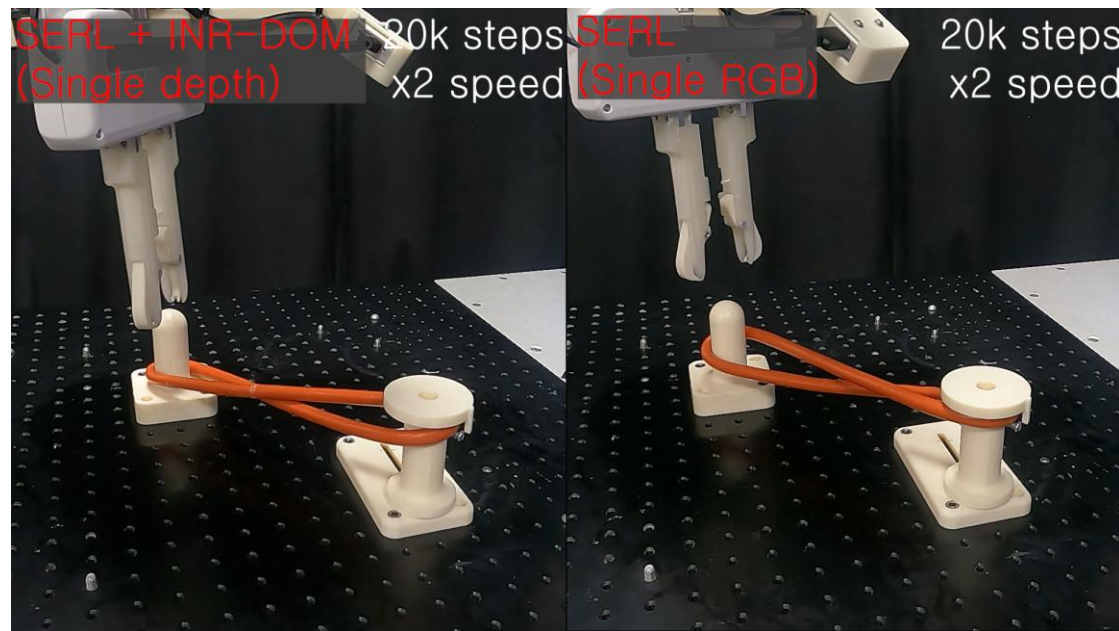
**Challenge:** Partially-observable unlimited DoF → sampling complexity in RL

**Solution:** Implicit neural representation learning with contrastive RL



Visually similar, but geometrically largely different DLOs

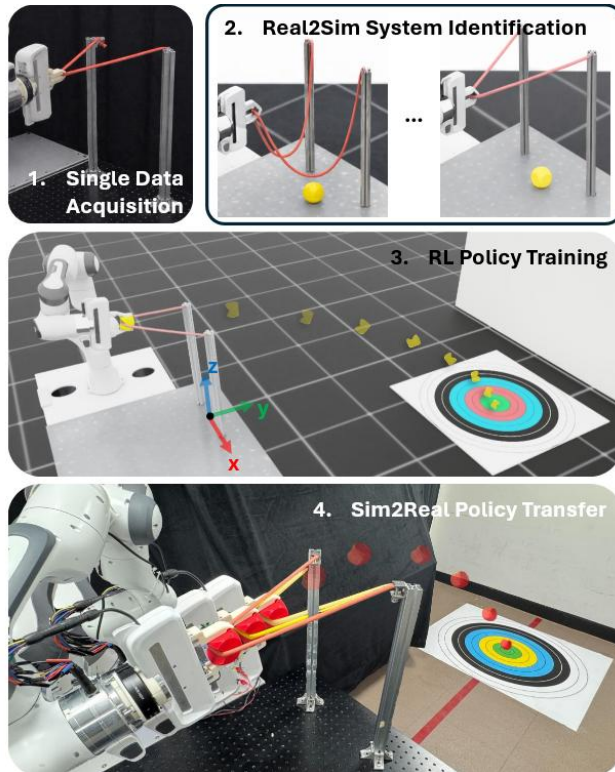
→ precise & consistent representations



# Sling2Sim2Real: One-Shot Elastic System Identification for Non-Destructive Slingshot Policy Learning

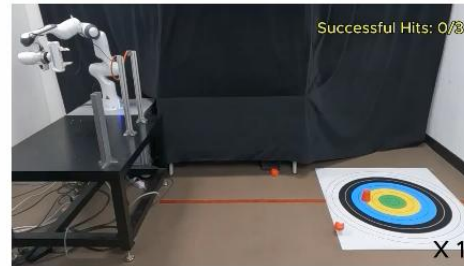
(IROS 2026)

Our **one-shot Real2Sim2Real framework** that identifies elastic parameters from a single non-destructive interaction and enables policy learning in simulation.

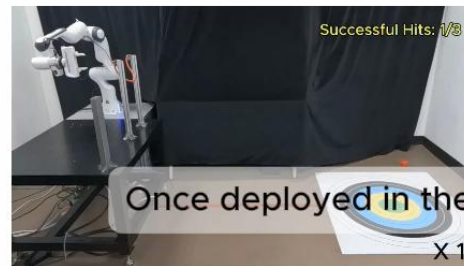


## Policy Transfer (Sim2Real)

### Real2Sim2Real (Baseline)

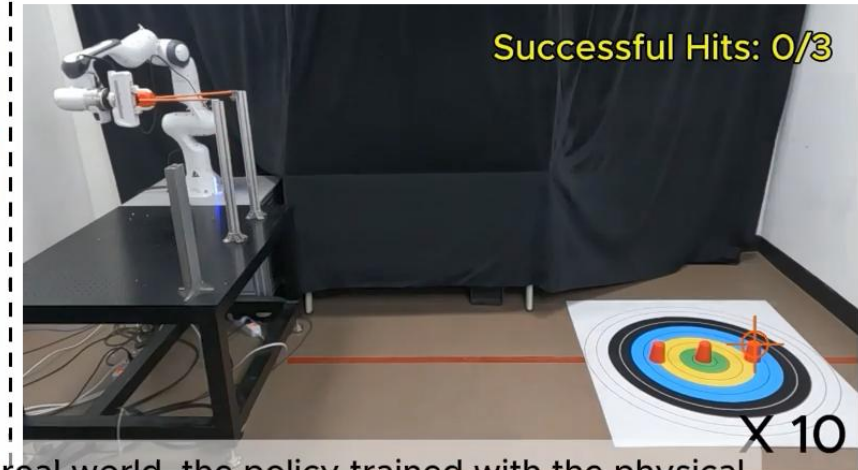


### Package SI (Baseline)



Once deployed in the real world, the policy trained with the physical

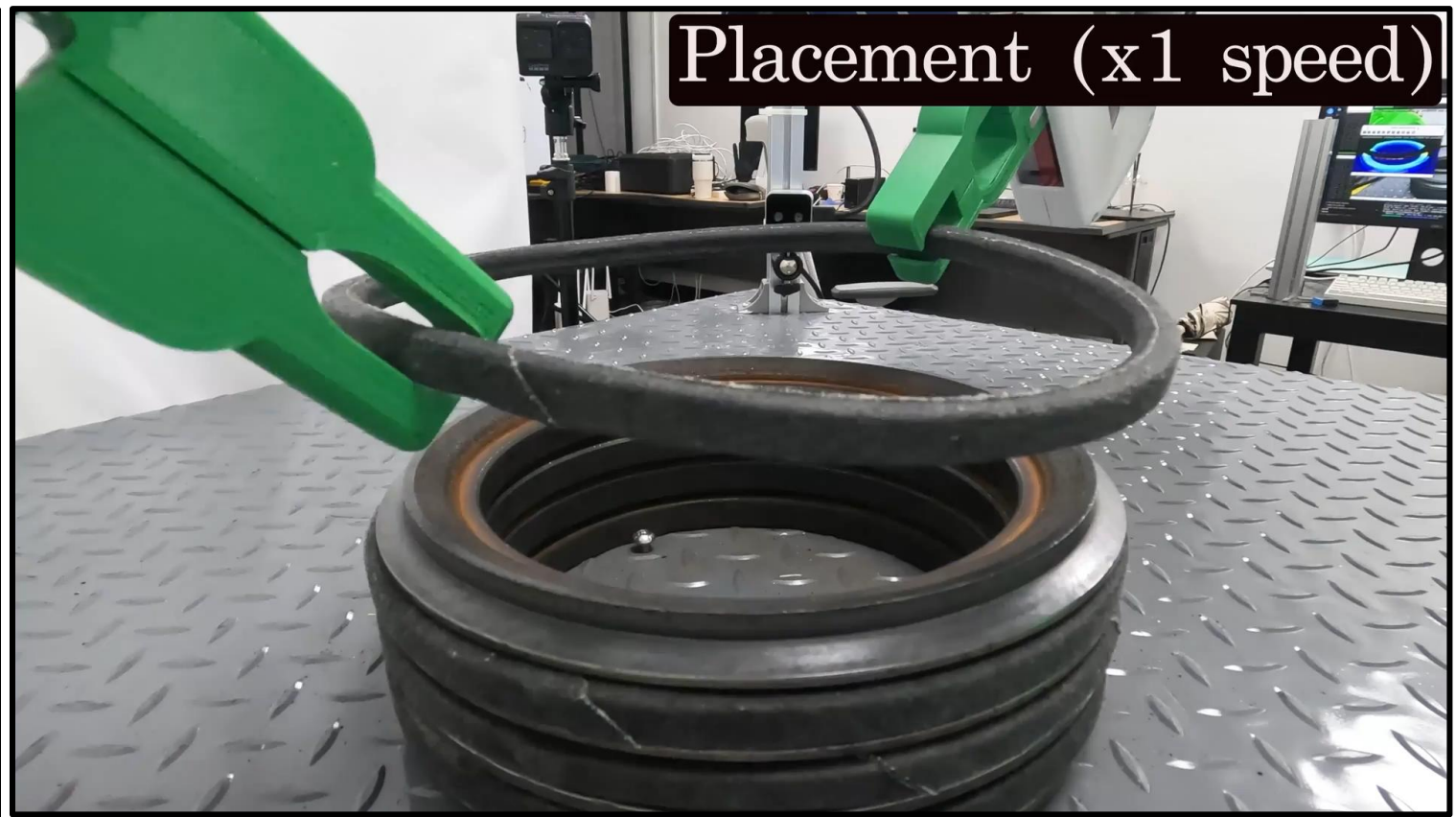
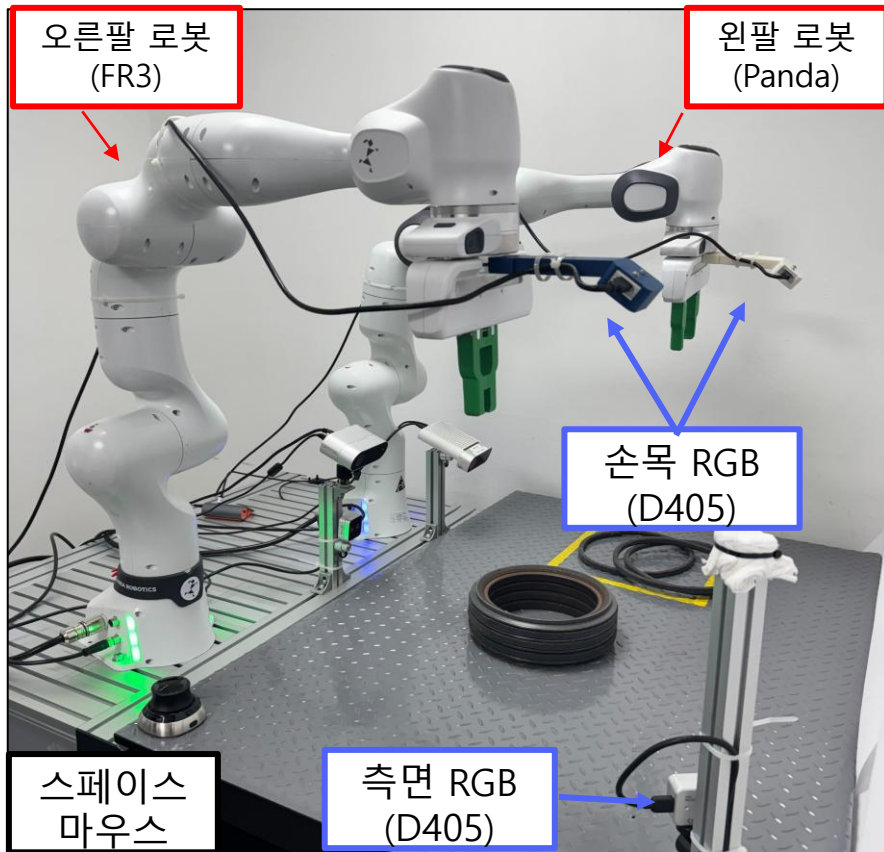
## Sling2Sim2Real (Ours)



# Heatmap-based Precise Sample-Efficient Robotic Reinforcement Learning for Rubber Compression Molding

(to be submitted)

We learn manipulation skills for highly elastic deformable objects using a dual arms via supervised reinforcement learning.



# Constraints in Everyday Tasks



Credit: Ljupco

## A wide **variety** of constraints

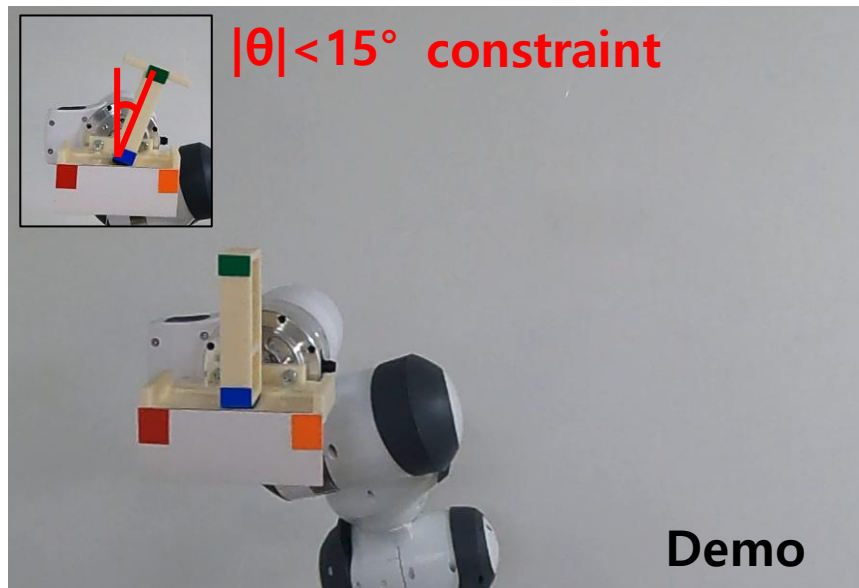
- Safety constraints
- Operational constraints
- Task-specific constraints
- Preference, legal, privacy, etc.

You may fail to complete the carrying without satisfying constraints.

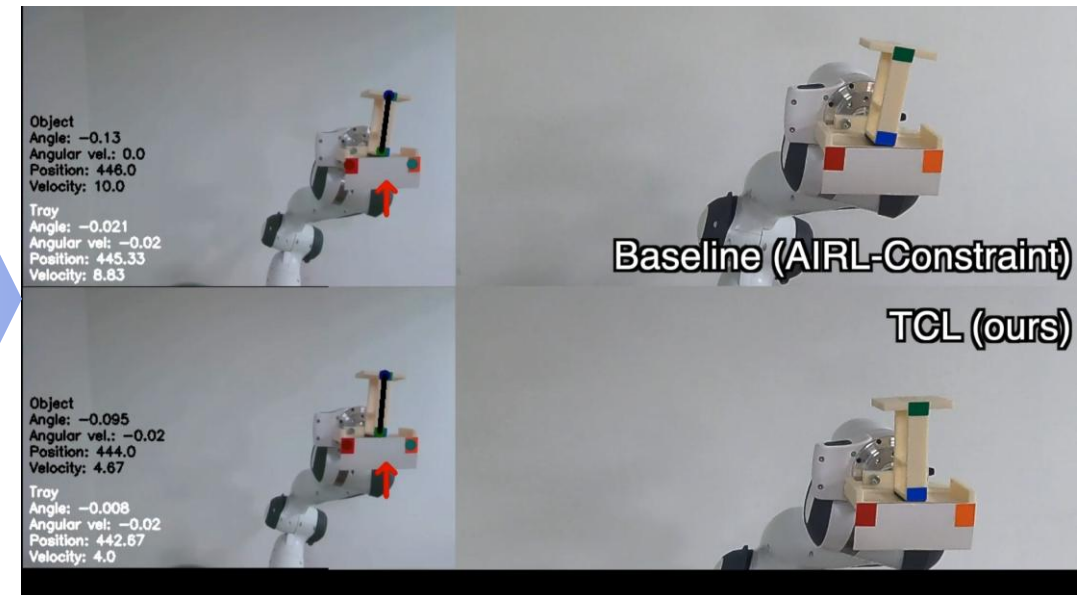
# Inverse Constraint Learning

## Transferable Constraint Learning (TCL)

A transferable constraint learning (TCL) algorithm that jointly infers a **task-relevant reward** and **task-agnostic constraint**.



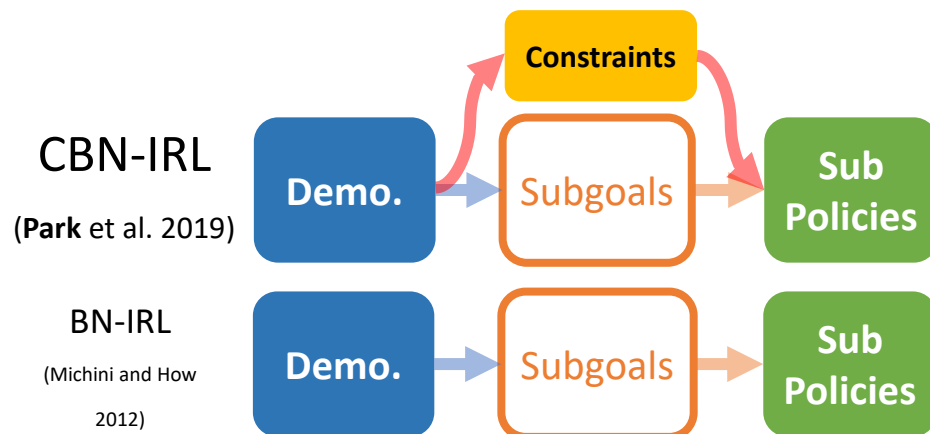
ICL+CRL



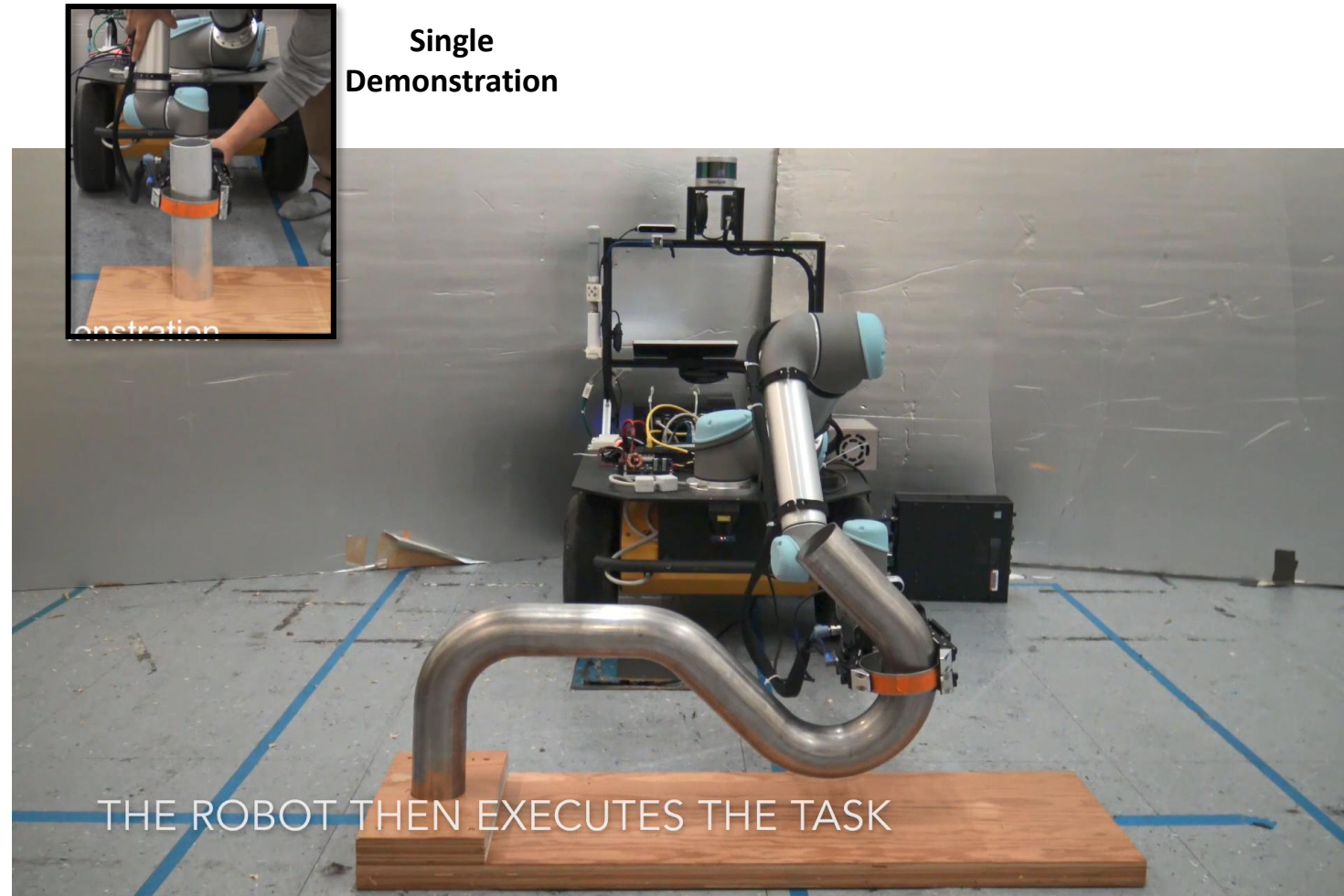
“

**Locally active constraints** are common in manipulation.

”



Single  
Demonstration



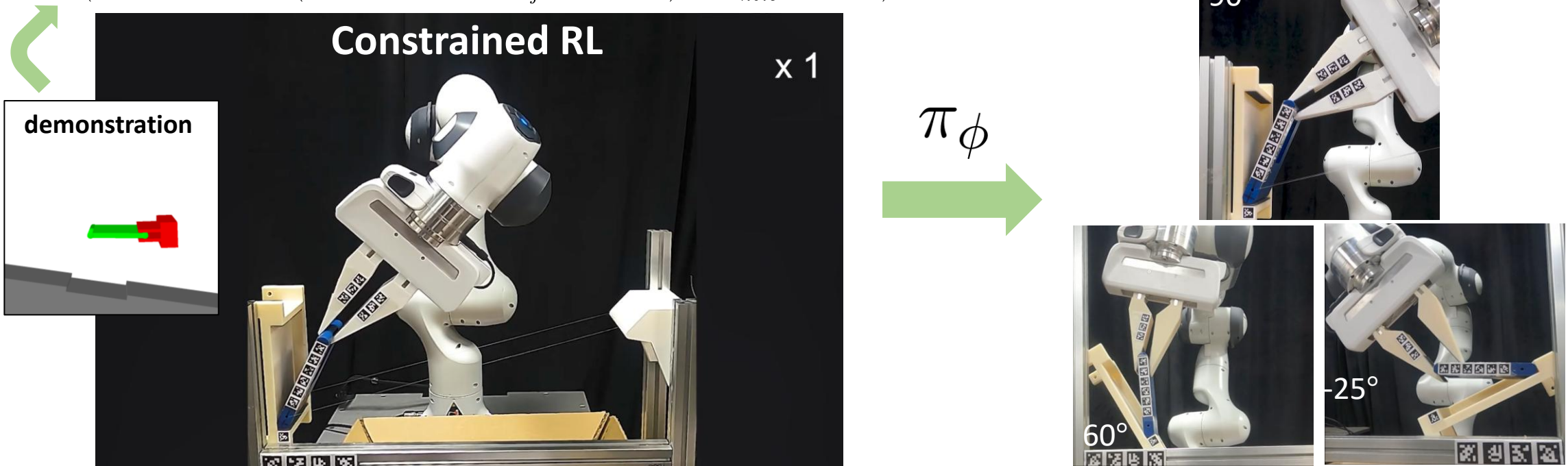
Park, Daehyung, et al. "Inferring task goals and constraints using bayesian nonparametric inverse reinforcement learning." CoRL, 2020.

# Inverse Constraint Learning

## ILCL : Inverse Logic-Constraint Learning

ILCL learns **temporal logic constraints** from temporally constrained demonstration leveraging two-player zero-sum game framework.

$$\diamond (\Sigma_h \theta^{peg} > 34.88^\circ \wedge (\Sigma_h \theta^{peg} < 4.217^\circ \mathcal{R} d_{jaw}^{peg} < 0.0053) \wedge \square d_{hole}^{peg} < 0.0072)$$



Cho, Minwoo, Jaehwi Jang, and Daehyung Park. "ILCL: Inverse Logic-Constraint Learning From Temporally Constrained Demonstrations." *RA-L*, 2025.

**Humans continuously negotiate and violate constraints imposed by both nature and society.**

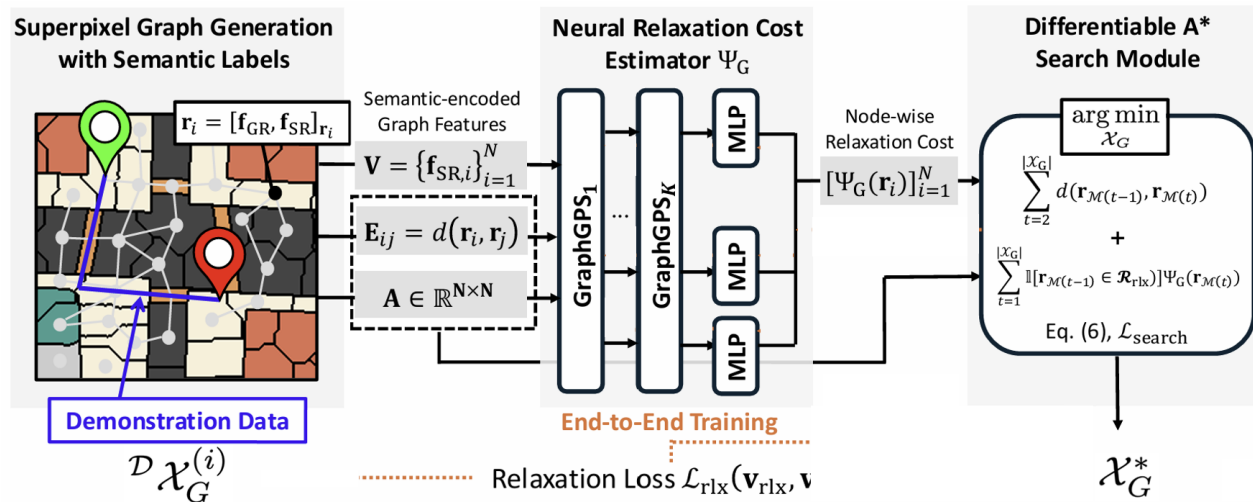


# SuReNav: Superpixel Graph-based Constraint Relaxation for Navigation in Over-constrained Environments

24

Can we find a best-effort solution that avoids all hard constraint regions while minimally traversing the least risky areas?

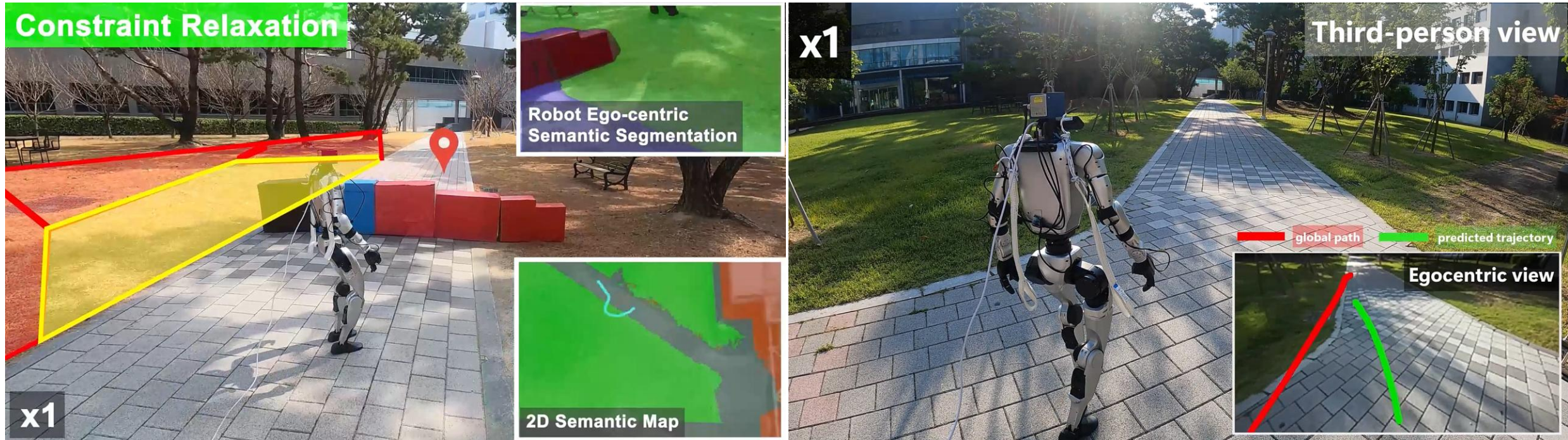
SuReNav learns to **automatically relaxes regional constraint** while planning a graph path.



# SuReNav<sup>+</sup>: Human-like Humanoid Nav.

(to be submitted)

We extend SuReNav by integrating **human preference-aware constraint relaxation** and navigation on a superpixel graph in and outdoor environments.

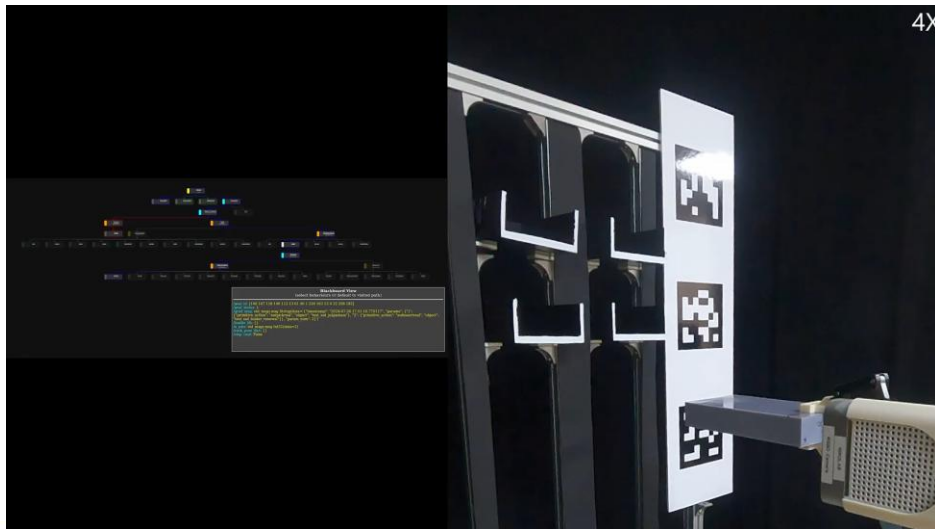


Visual local planner

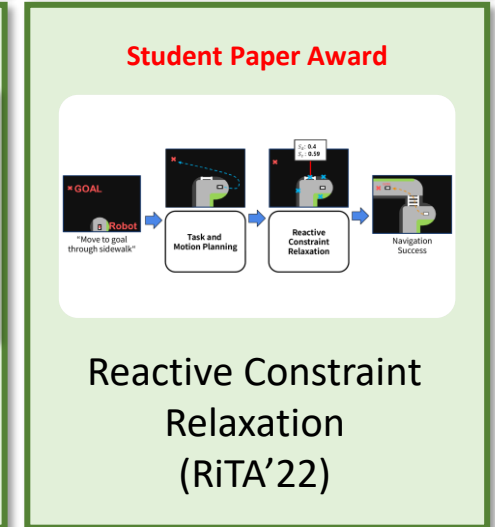
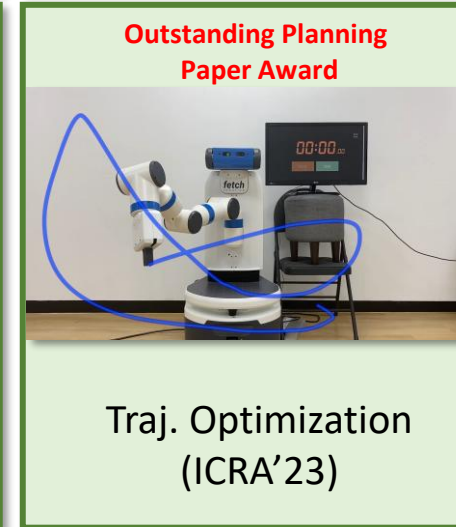
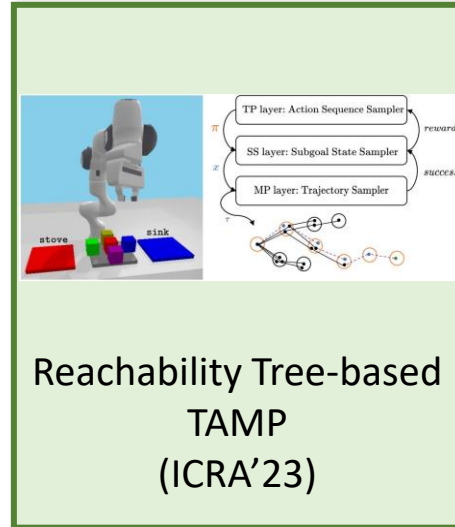
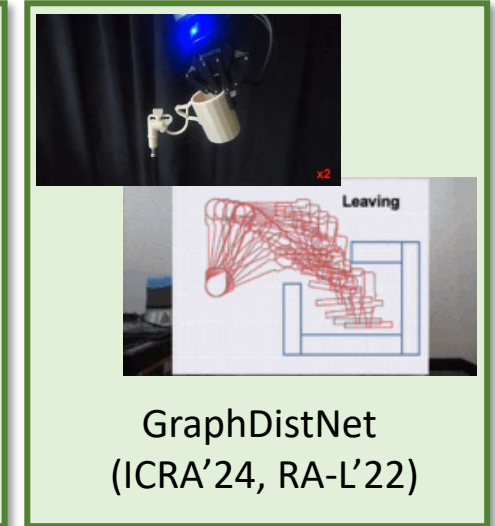
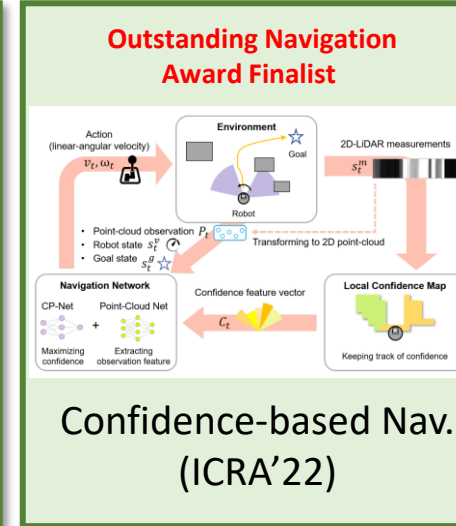
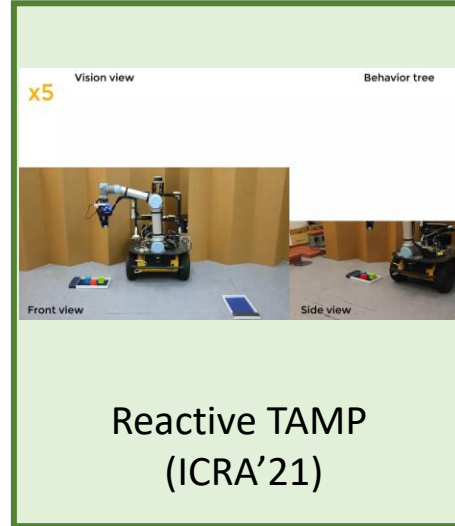


# Task and/or Motion Planning

- Motion planning under environmental changes or occlusions
- Task (re)-planning under changes
- Online-recovery with behavior tree
- Learning-based collision check



Recovery with behavior tree

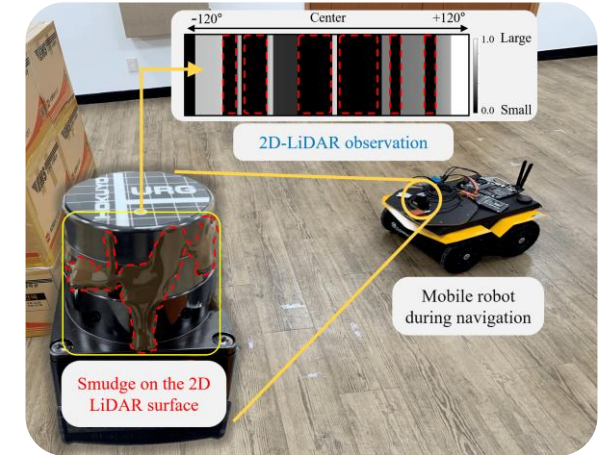
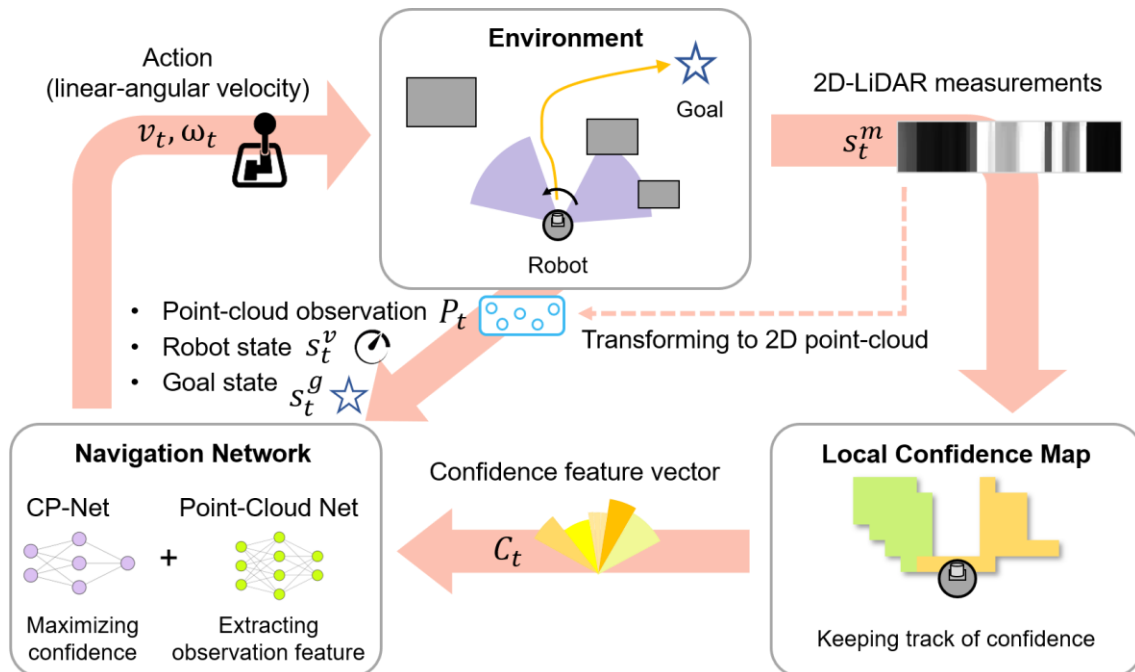


# Confidence-Based Robot Navigation Under Sensor Occlusion

ICRA22 – **Outstanding  
Navigation Paper Finalist**

Consider unexpected occlusions on navigation sensors.

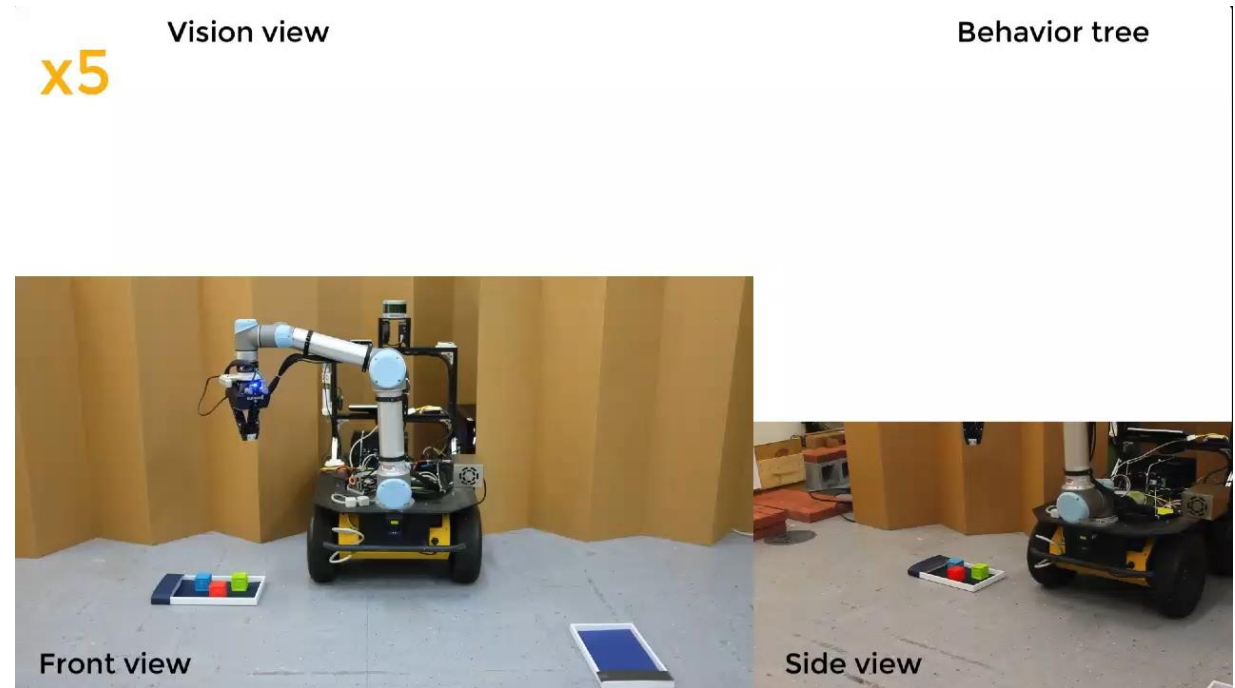
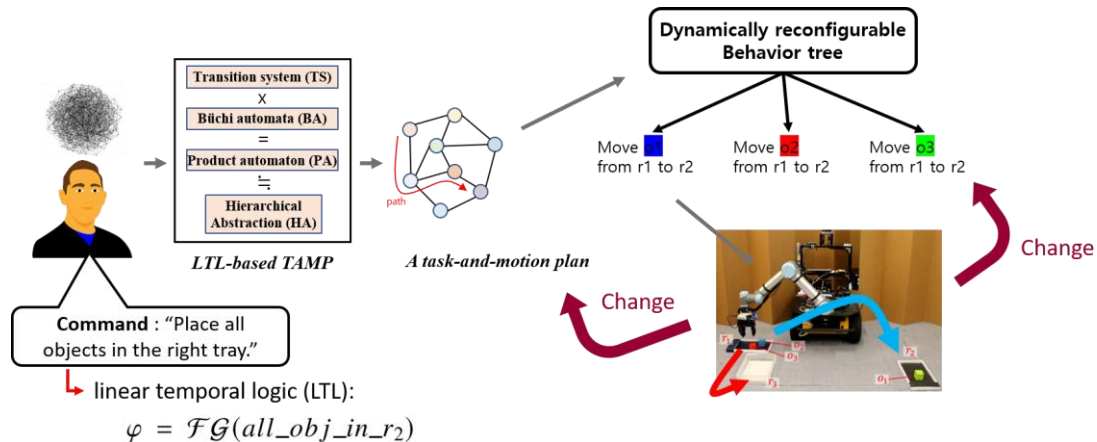
We propose a **confidence-based navigation method** trained through DRL. (Collaboration with Prof. Sung-eui Yoon)



# Task and Motion Planning under Changes

We propose **TAMP with Linear Temporal Logic (LTL)**. (Collaboration with Dr. Shen Li and Dr. Yoonchang Sung)

The hierarchical architecture for efficient replanning and execution to handle environmental changes



# ForeSight: Autoregressive Plan Monitoring for Efficient Long-Horizon Replanning and Execution



Yeseung



Gyungjin

(under review)

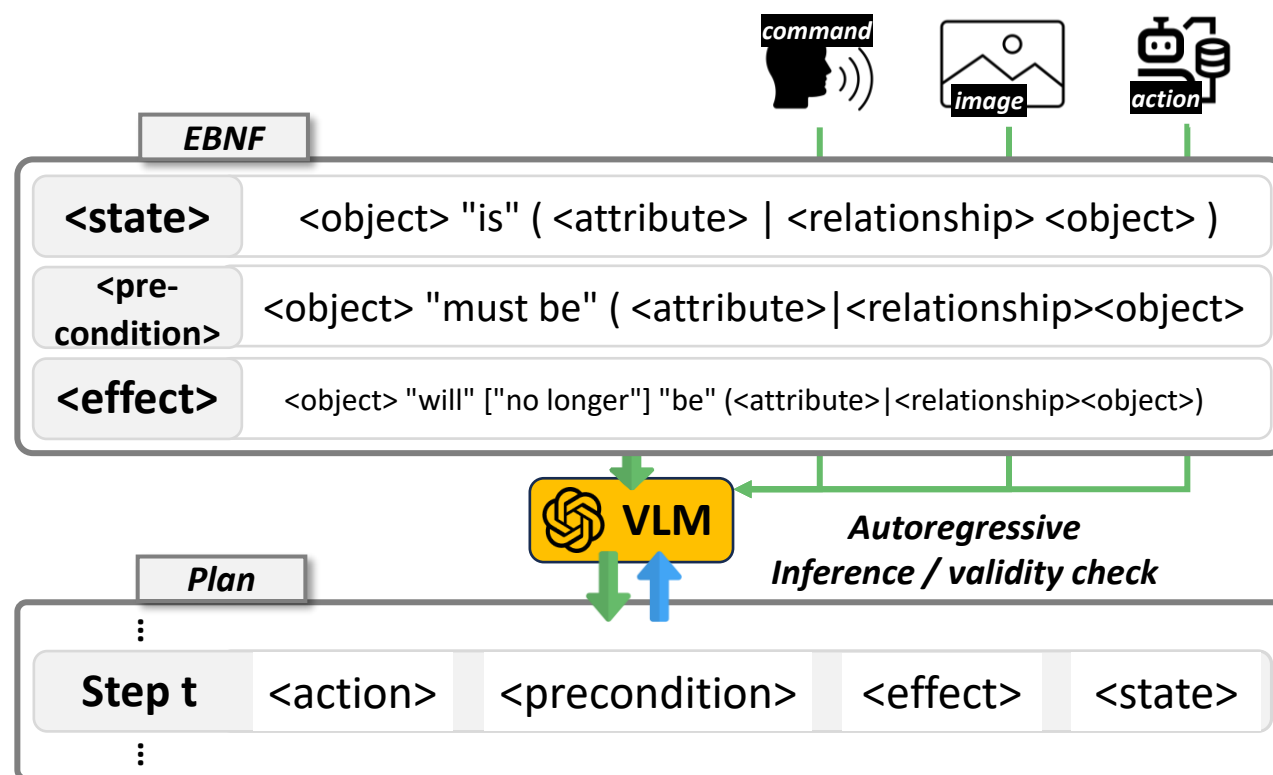
- **Extended Backus–Naur Form (EBNF)-based Chain-of-State prompt**

→ *Consistent representation for long-horizon task planning*

- **Autoregressive plan validity checking**

→ *Early detection + Reactive planning*

Task: Build a tower using cubes and one bar.

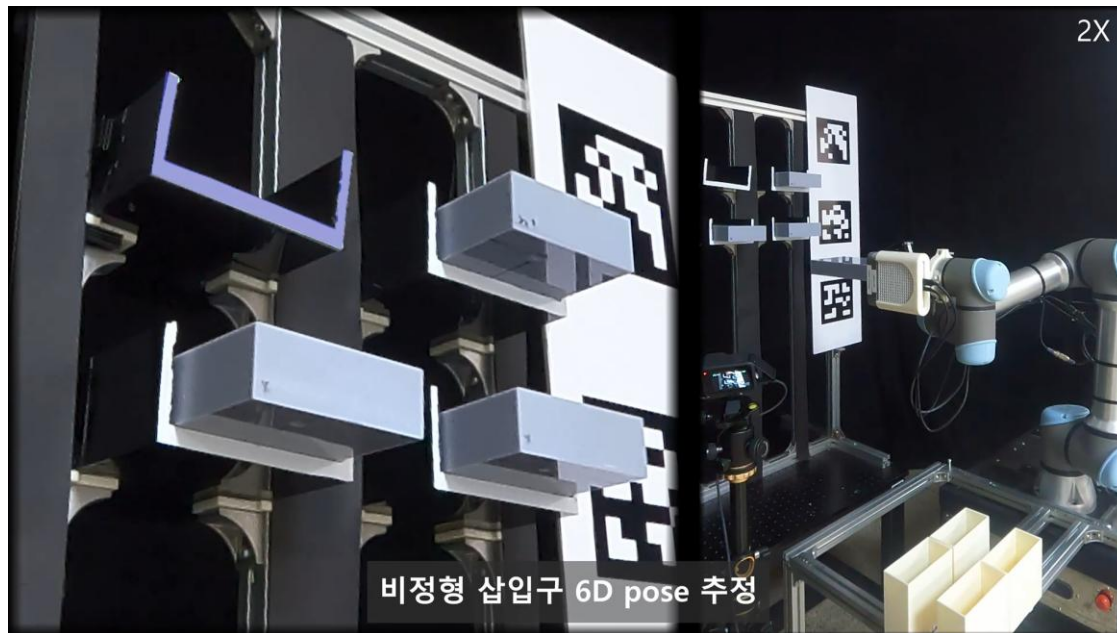


# Multimodal Task-and-Motion Planning for multi-pack insertation task

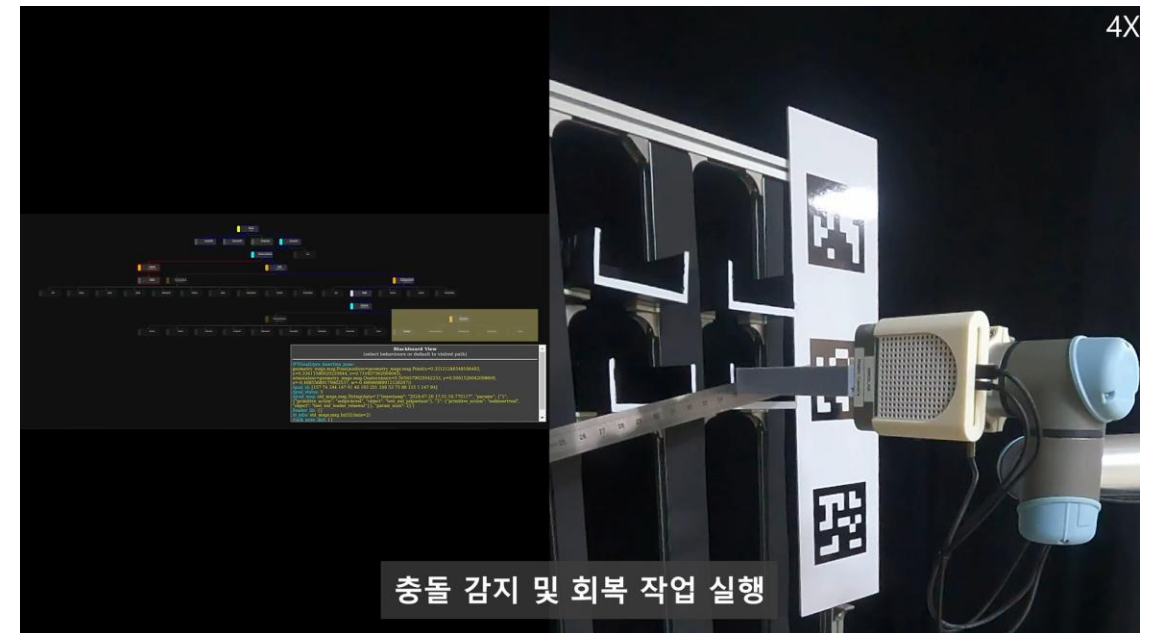
**SEMES**

## Task planning and multi-modal recovery using behavior tree

- 6D pose estimation for multi holes in real-world
- Safe motion planning and simulation



Automatic HDD/SDD quality check system



Recovery with behavior tree

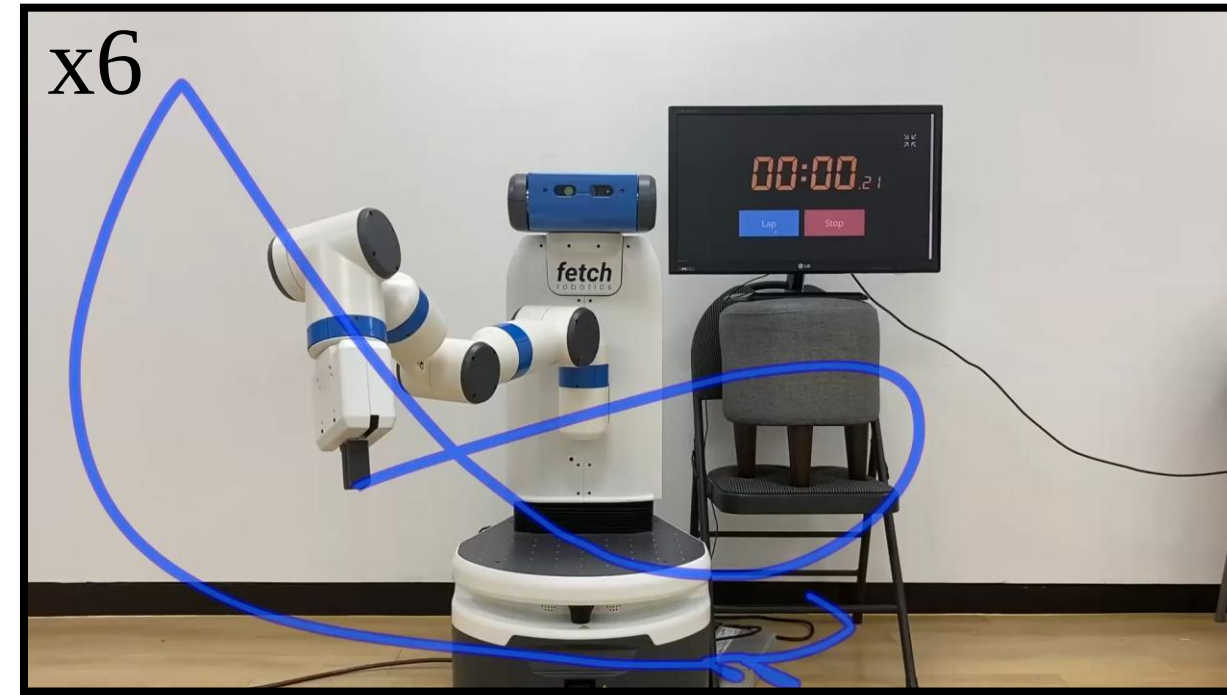
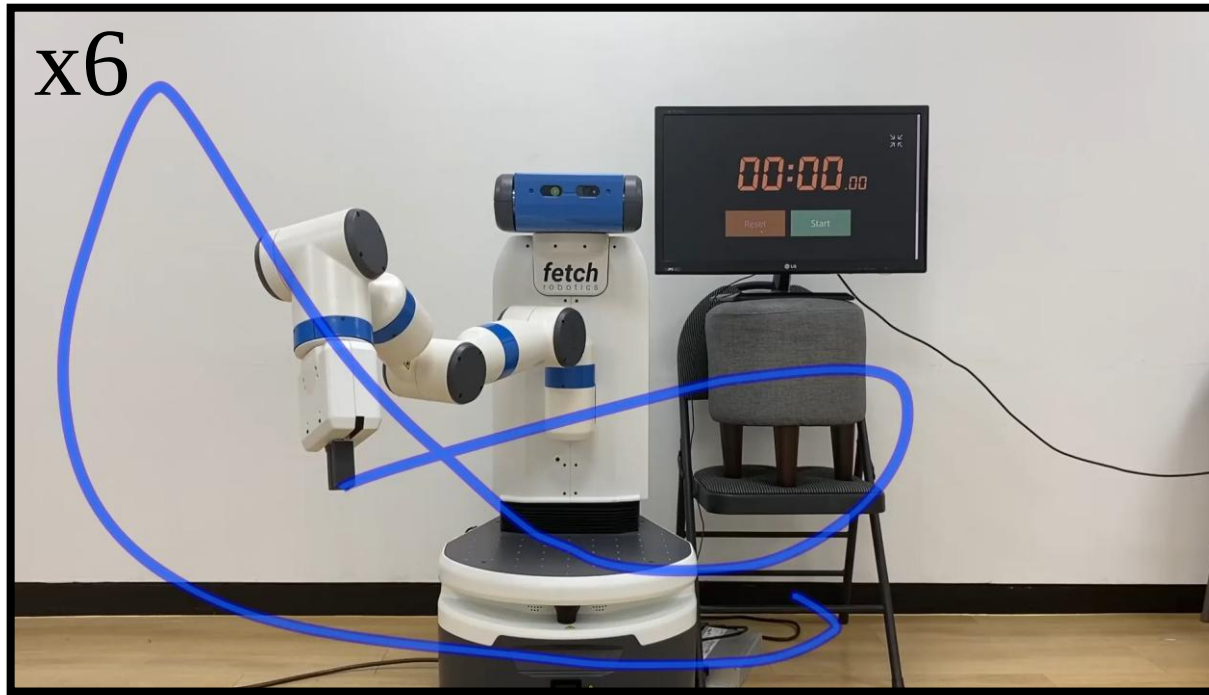
# Learning-based Initialization of Trajectory Optimization

ICRA23 – **Outstanding  
Planning Paper Award**

- Benchmark name: 'Random #64' | Trajectory optimizer: TORM [Kang et al. 2020]

*with **RL-ITG** (ours)*

*with **Greedy** (baseline)*



- Total execution time: **00:46.76**

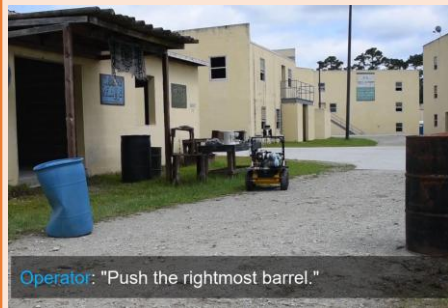
*about **220%** faster than **Greedy***

**01:43.16**

min : sec



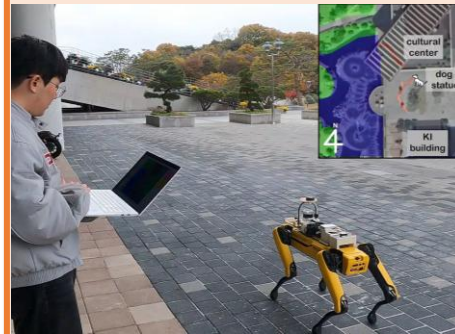
# Natural-Language Grounding



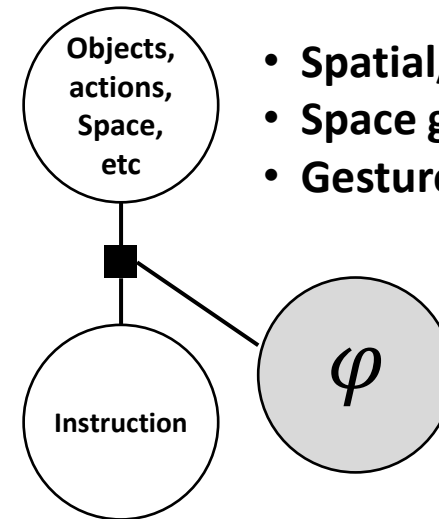
Multi-modal Estimation  
& Communication  
(IJRR'20)



LINGO-Space for  
Language Grounding  
(AAAI'24)



Language-guided  
Navigation  
(Proj. 23-24)



- Spatial/space grounding
- Space grounding-based VLA
- Gesture grounding



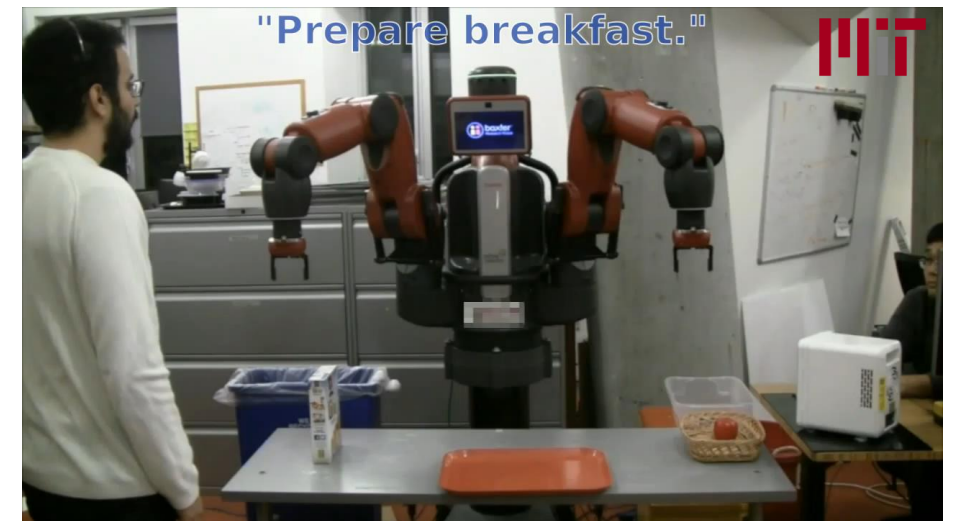
SGGNet2: Speech-  
guided Navigation  
(RO-MAN'23)



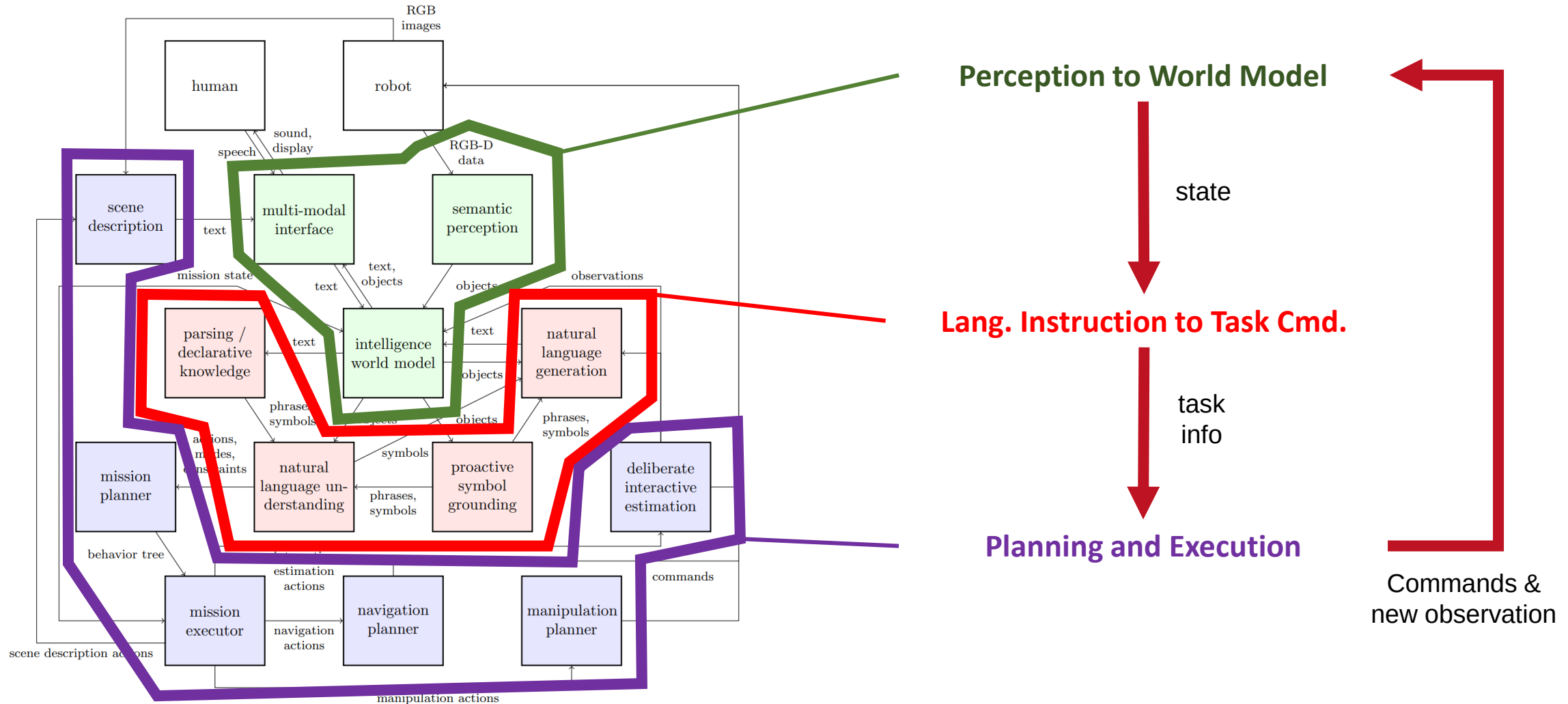
Grounding Robot Plans  
(CoRL'18)



Model-free space  
grounding (ISR'26)



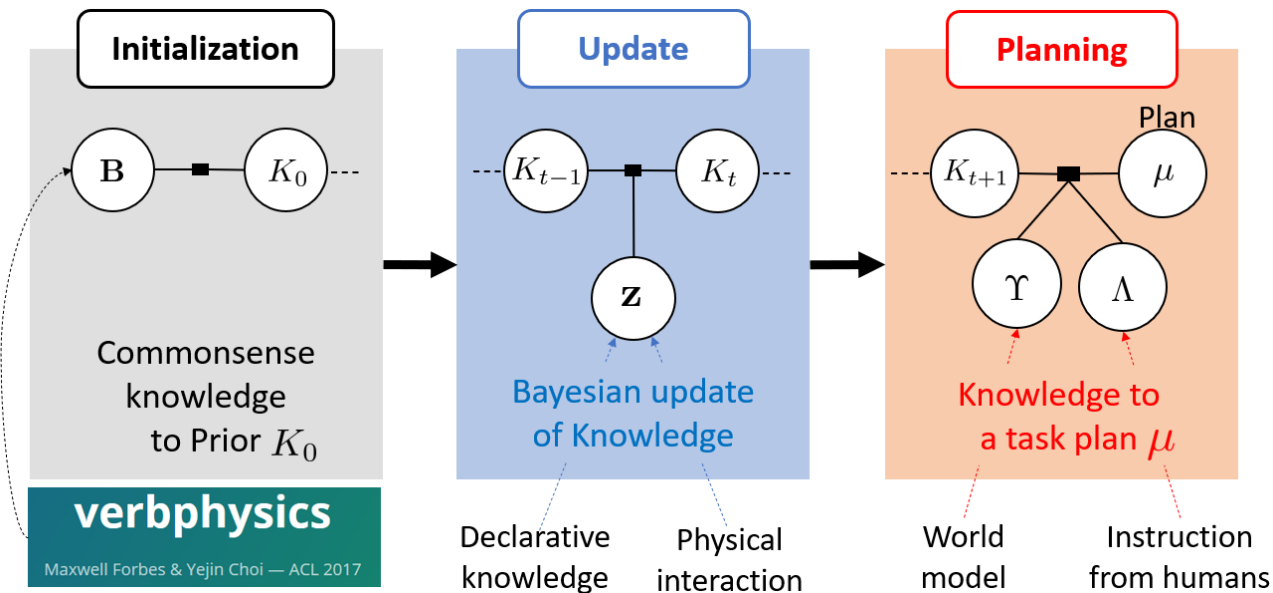
# Intelligence Architecture for Grounded Language

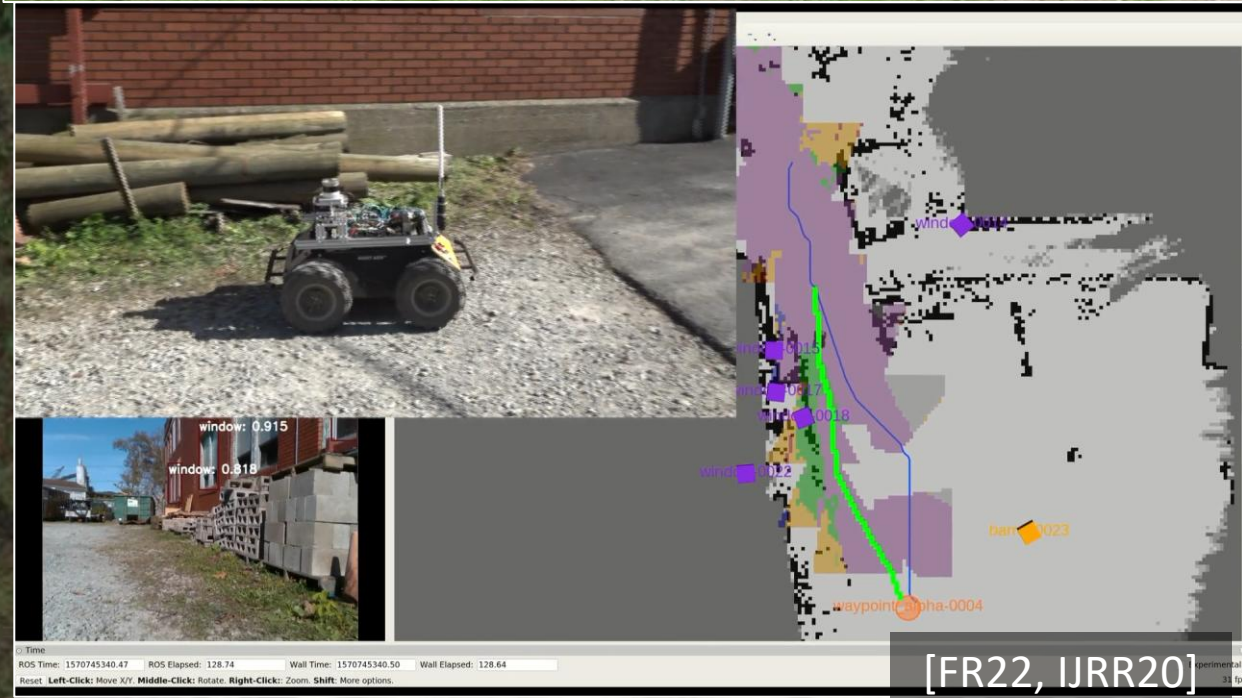


# Multimodal Estimation and Communication

## Natural language-driven mobile manipulation

- Infer semantic knowledge of the world using a Bayesian framework
- Correct invalid knowledge and generate a safe plan





[FR22, IJRR20]

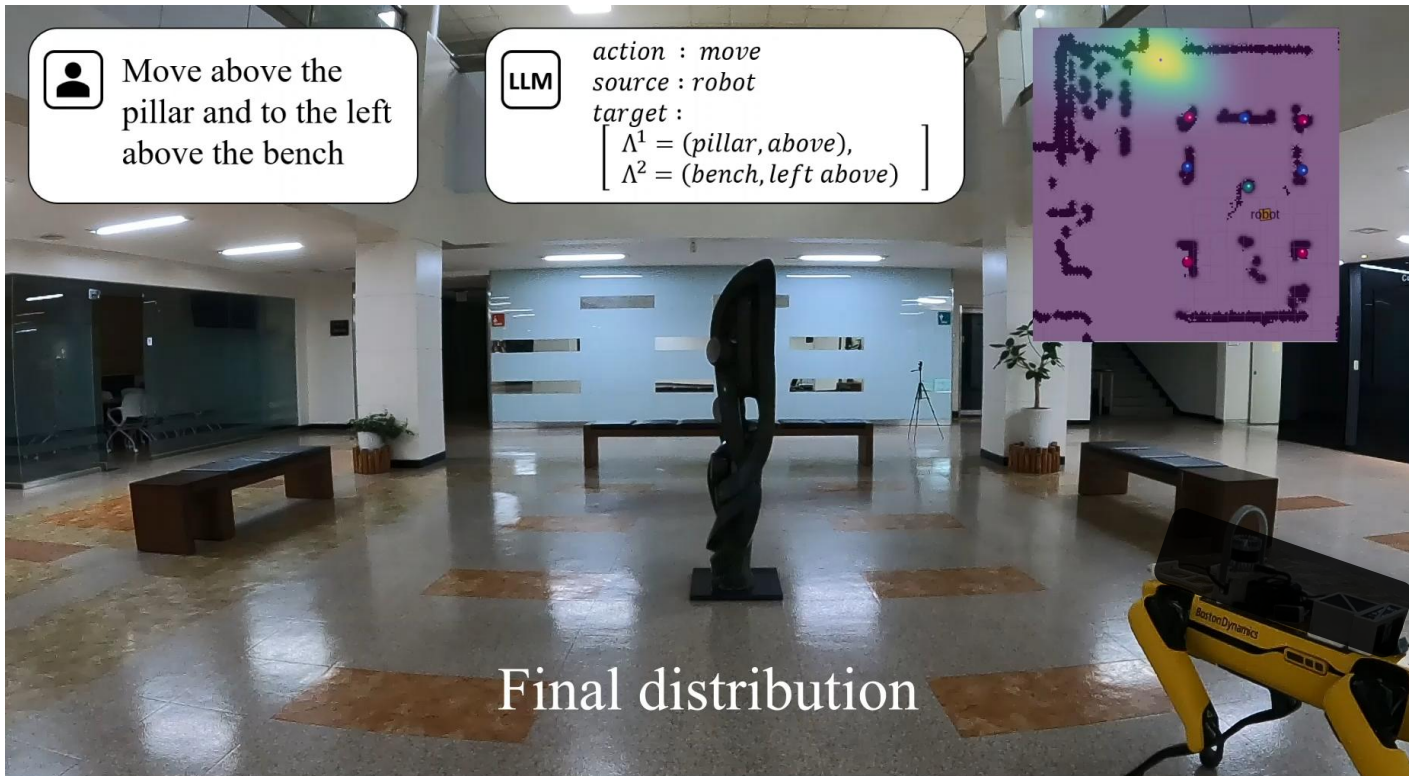
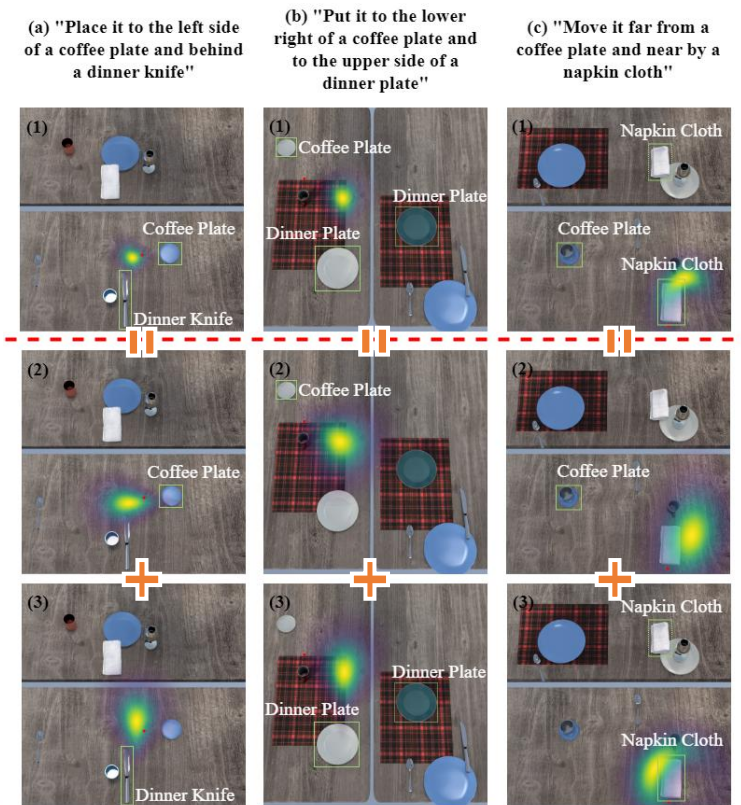


Dohyun

# Model-based Space Grounding

## LINGO-Space: Language-Conditioned Incremental Grounding

- Our mixture of polar distributions represents positions from combined instructions.
- **The probabilistic grounding model** infers multiple placement candidates !!



# Model-free Space Grounding



Dohyun



Junhyeong

37

(ISR 2026)

## C2F-SPACE: Coarse-to-Fine Space Grounding for Spatial Instructions using Vision-Language Models

(Collaboration with Dr. Rohan Paul, IIT Delhi)



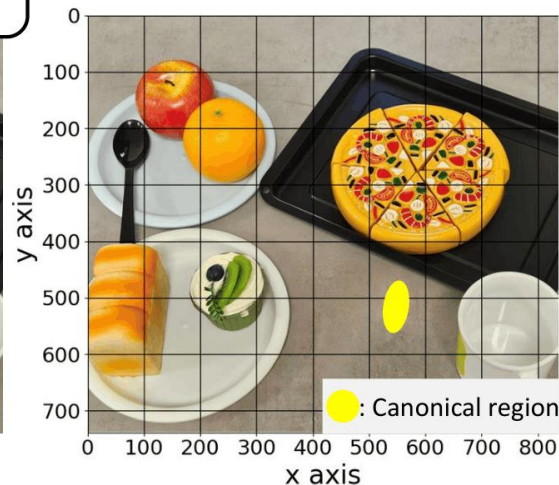
KAIST

- Estimates an approximated yet spatially consistent region using a VLM.
- Refines the region to align with the local environment through superpixelization

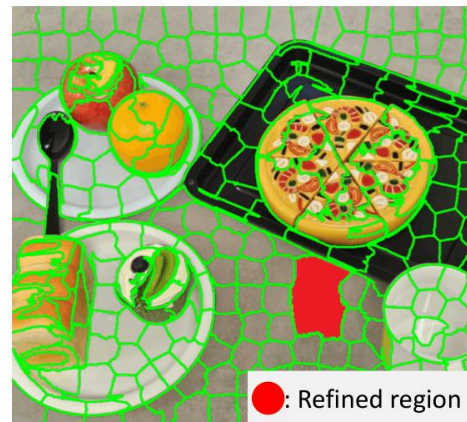
Place the spoon to the **right** of the cupcake at **twice** the distance between the cup and the pizza.



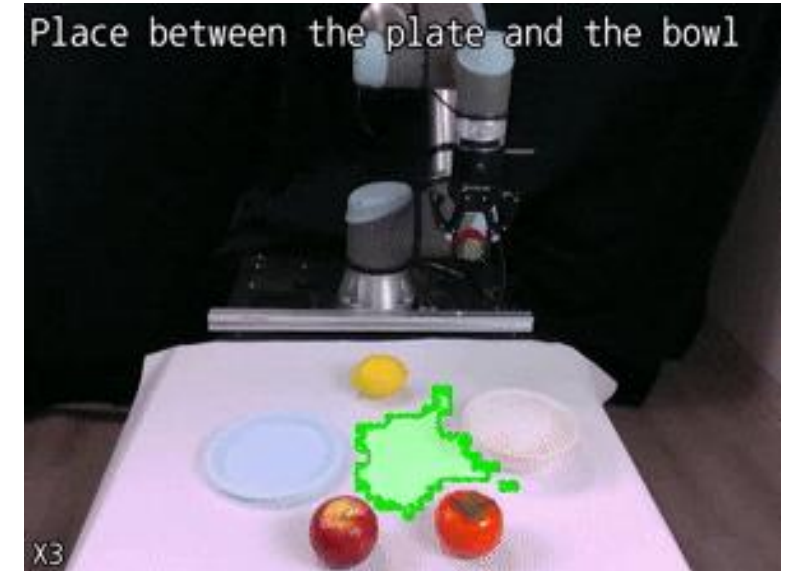
**Stage 1:** Space reasoning with physical & semantic validators



**Stage 2:** Superpixel based space refinement



Two stage grounding



Pixel-grounding with robot manipulation

# Space Grounding at Pixel Level using Attentions of Vision Language Models



Dohyun

(to be submitted)

We infer referred space by leveraging the inherent spatial reasoning capabilities of **off-the-shelf VLMs** through their attention maps.

X2.5



"vacant area between the green plate and blue plate"



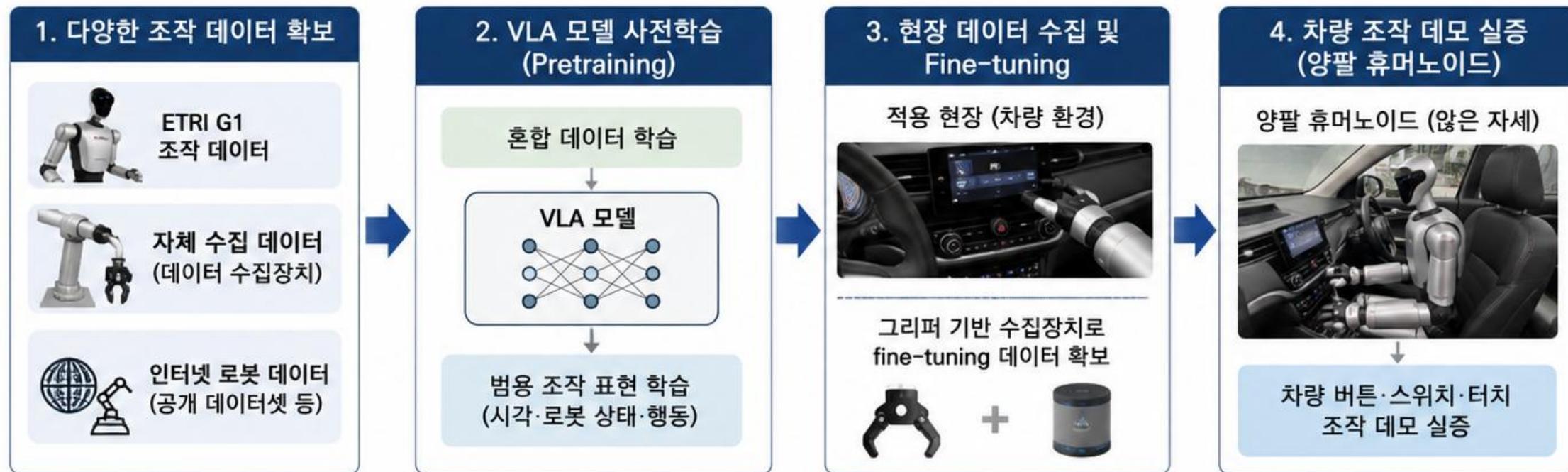
"vacant area between the plate and the bowls"



"vacant area between gray plate, blue plate, and blue bowl"

# Whole-body Manipulation

- Physically-constrained whole-body motion generation
- Bimanual manipulation (VLA + RFT) for industrial / assistive tasks

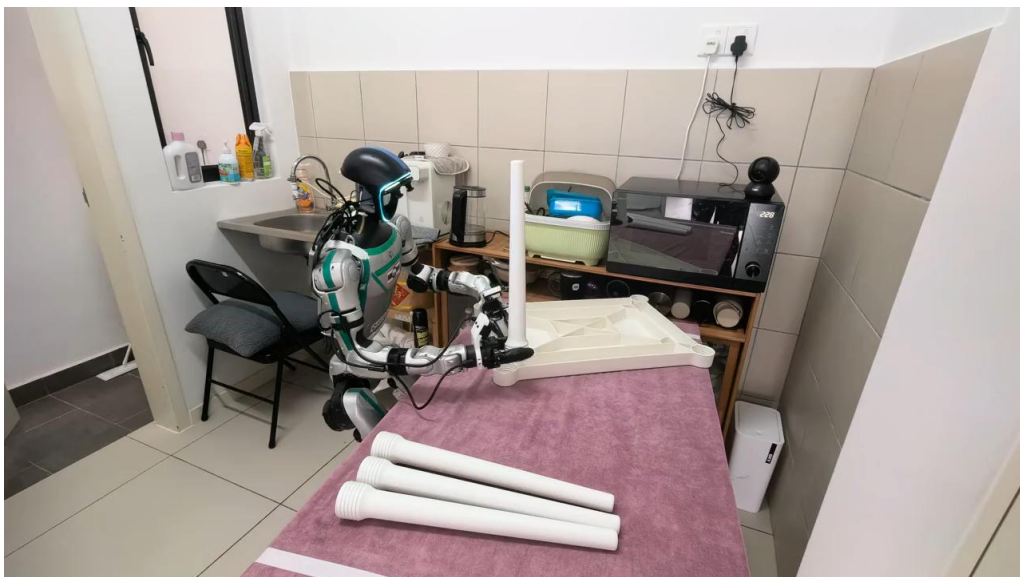


# Whole-body Manipulation

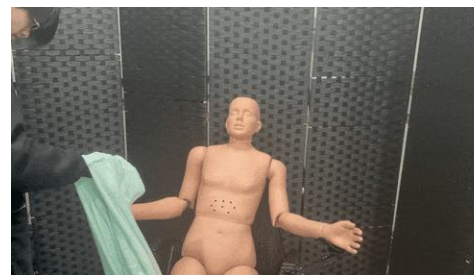
- Physically-constrained whole-body motion generation
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Humanoid IKEA Assembly Challenge



PhyRC Challenge



Co-organization with Cornell Univ.

# Thank you. Any questions?



Plz, check our website [rirolab.kaist.ac.kr](http://rirolab.kaist.ac.kr) or send an email to [daehyung@kaist.ac.kr](mailto:daehyung@kaist.ac.kr) !